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FURTHER MATHEMATICS • UPPER SIXTH

UPPER SIXTH — FURTHER MATHEMATICS

## CHAPTER 17

# DISCRETE RANDOM VARIABLES

SUGGESTED TIME: 6 PERIODS

IN THIS CHAPTER

This chapter introduces discrete random variables and their probability distributions, the fundamental language of probability at Advanced Level. Students learn to construct and use probability mass functions and cumulative distribution functions, compute and interpret expectation, variance, mode and median, and discover moment generating functions as a powerful tool for analysing distributions.

- By the end of this chapter I can define a discrete random variable and distinguish it from a continuous one.
- By the end of this chapter I can construct the probability mass function (p.m.f) of a discrete random variable from a described experiment and present it as a table or formula.
- By the end of this chapter I can verify and use the two conditions for a valid p.m.f (non-negativity and total probability equal to 1), including finding unknown constants.
- By the end of this chapter I can compute the expectation E(X) and the expectation of a function E[g(X)], and apply the linearity properties of expectation.
- By the end of this chapter I can compute the variance Var(X) using $\text{Var}(X) = E(X^2) - [E(X)]^2$ and apply the properties $\text{Var}(aX + b) = a^2\text{Var}(X)$.
- By the end of this chapter I can construct and use the cumulative distribution function $F(x) = P(X \leq x)$, and recover the p.m.f from it.


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2. The probability mass function and expectation
3. Variance and standard deviation of a discrete random variable
4. The cumulative distribution function, mode and median
5. Moment generating functions
6. Integration activity
7. Methods and worked examples
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CHAPTER OBJECTIVES

- By the end of this chapter I can define a discrete random variable and distinguish it from a continuous one.
- By the end of this chapter I can construct the probability mass function (p.m.f) of a discrete random variable from a described experiment and present it as a table or formula.
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- By the end of this chapter I can construct and use the cumulative distribution function $F(x) = P(X \leq x)$, and recover the p.m.f from it.
- By the end of this chapter I can determine the mode and the median of a discrete random variable from its distribution.
- By the end of this chapter I can define the moment generating function M(t) = E(e^(tX)), compute it for simple distributions, and use it to obtain E(X) and Var(X).
- By the end of this chapter I can model simple real-life situations (games of chance, counts of events) with a discrete random variable and interpret E(X) and Var(X) in context.

PREREQUISITES

- Elementary probability from Lower Sixth: sample spaces, events, P(A), addition and multiplication rules, conditional probability and independence.
- Sigma notation and manipulation of finite sums.
- Basic algebra: solving linear and quadratic equations, manipulating indices and the exponential function e^x.
- Functions and their graphs (Lower Sixth pure mathematics), including piecewise-defined functions.
- Simple differentiation of polynomials and exponentials (needed for moment generating functions).

# 1 DISCRETE RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS

At the Mokolo market in Yaoundé, a trader selling phone credit cards never knows in advance how many customers will come each day: some days 3, some days 27. Yet she must decide every morning how many cards to stock, because unsold cards tie up her money while shortages lose her customers. How can she summarise this uncertainty with a single table, compute the 'average' number of customers to expect, and measure how wildly the demand fluctuates? The mathematics of **discrete random variables** — probability distributions, expectation, variance, cumulative functions, modes, medians and moment generating functions — gives her exactly the tools she needs, and gives you the tools the GCE Advanced Level examiners will demand. In this first section we build the foundation: what a random variable is, and how its probability distribution is defined, displayed and used.

## 1.1 Random experiments revisited

A **random experiment** is any process whose outcome cannot be predicted with certainty in advance, even though the set of all possible outcomes is known. Tossing a coin, rolling a die, counting the number of customers at the Mokolo stall in one day: all are random experiments.

Recall from Lower Sixth probability:

- The **sample space** $\Omega$ is the set of all possible outcomes of the experiment.
- An **outcome** $\omega$ is a single element of $\Omega$.
- An **event** $A$ is a subset of $\Omega$; its probability $P(A)$ satisfies $0 \le P(A) \le 1$, with $P(\Omega) = 1$.

For example, when two fair coins are tossed, $\Omega = \{HH, HT, TH, TT\}$, and since the coins are fair, each outcome has probability $\frac{1}{4}$.

The difficulty is that outcomes like $HT$ or "rain in Bamenda tomorrow" are not numbers. To do serious mathematics — to average, to compare, to optimise — we must attach numbers to outcomes. This is exactly what a random variable does.

## 1.2 What is a random variable?

**DEFINITION** **Random variable**

A **random variable** $X$ on a sample space $\Omega$ is a function

$$X : \Omega \longrightarrow \mathbb{R}, \qquad \omega \longmapsto X(\omega),$$

which assigns to each outcome $\omega$ of a random experiment a real number $X(\omega)$.

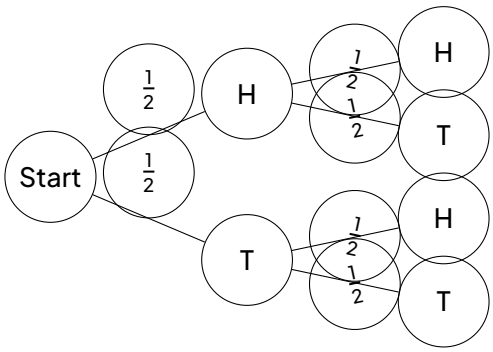
The key word is function. A random variable is neither random nor a variable in the algebraic sense: it is a perfectly deterministic rule that converts outcomes into numbers. The randomness lies in the experiment, not in the rule.

**EXEMPLE** **Number of heads in two coin tosses**

Toss two fair coins, so $\Omega = \{HH, HT, TH, TT\}$. Define $X$ = "number of heads obtained". Then

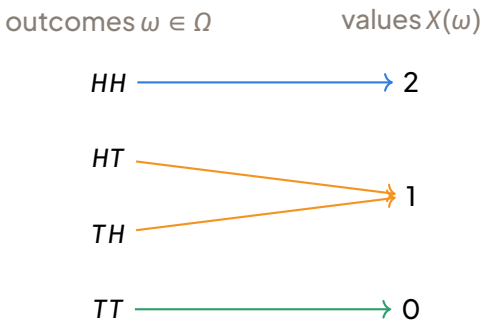
$$X(HH) = 2, \quad X(HT) = 1, \quad X(TH) = 1, \quad X(TT) = 0.$$

Notice that two different outcomes, $HT$ and $TH$, are mapped to the same value 1. A random variable need not be one-to-one.




Tree diagram for two fair coins. The four final nodes represent $TT, TH, HT, HH$ (left to right).

The mapping is clearer when drawn explicitly:




The random variable $X$ as a function from $\Omega$ to $\mathbb{R}$: $HT$ and $TH$ both map to 1.

## 1.3 Discrete versus continuous random variables

**DEFINITION** **Discrete random variable**

A random variable $X$ is **discrete** if it takes only finitely many values or countably many val-

ues (that is, values that can be listed $x_1, x_2, x_3, \ldots$, like the natural numbers).

By contrast, a **continuous** random variable can take any value in an interval of $\mathbb{R}$; it is studied in a later chapter.

**EXEMPLE** **Discrete or continuous? Cameroonian situations**

- The number of goals scored by the Indomitable Lions in one match: $0, 1, 2, \ldots$ — discrete.
- The number of students passing the GCE in a school in Buea: $0, 1, 2, \ldots$ — discrete.
- The number of customers at the Mokolo stall in one day — discrete.
- The height of a randomly chosen Upper Sixth student, which could be 1.72 m, 1.723 m, 1.7234 m, … — continuous.
- The time a bus takes from Douala to Yaoundé — continuous.

The test is simple: if you count, the variable is discrete; if you measure, it is continuous.

## 1.4 Notation: the variable $X$ and the event $\{X = x\}$

We always use a **capital letter** $X$ (or $Y$, $T$, …) for the random variable itself — the function — and a **small letter** $x$ for a particular numerical value it may take.

Because $X$ is a function on $\Omega$, the statement "$X$ takes the value $x$" describes an event:

$$\{X = x\} = \{\omega \in \Omega : X(\omega) = x\}.$$

Its probability is written $P(X = x)$. For the two coins:

$$P(X = 1) = P(\{HT, TH\}) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

**WARNING** **Do not confuse $x$ with $P(X = x)$**

A very common mistake at GCE is to mix up the value $x$ of the variable with its probability $P(X = x)$. The value $x$ is a number the variable can take (e.g. 2 heads); the probability $P(X = x)$ is a number between 0 and 1 telling how likely that value is. In a distribution table, the first row contains values, the second row contains probabilities — never the reverse.

## 1.5 The probability distribution of a discrete random variable

**DEFINITION** **Probability distribution and p.m.f.**

Let $X$ be a discrete random variable taking values $x_1, x_2, \ldots$. The **probability distribution** of $X$ is the set of all pairs

$$\big(x_i,\ P(X = x_i)\big).$$

The function $p$ defined by $p(x) = P(X = x)$ is called the **probability mass function** (p.m.f.) of $X$.

A distribution can be presented in three equivalent ways.

**1. As a table.** For $X$ = number of heads in two coin tosses:

| $x$ | 0 | 1 | 2 |
|---|---|---|---|
| $P(X = x)$ | $\frac{1}{4}$ | $\frac{1}{2}$ | $\frac{1}{4}$ |

**2. As a formula.** The same distribution is given by

$$P(X = x) = \binom{2}{x}\left(\tfrac{1}{2}\right)^2, \qquad x = 0, 1, 2.$$

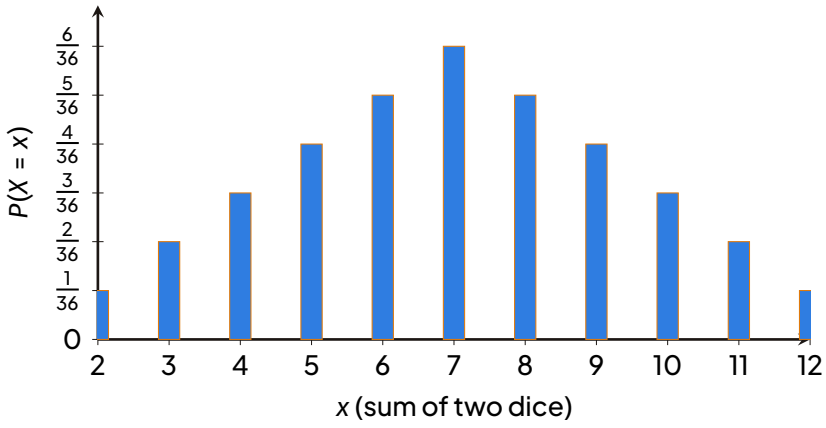
**3. As a bar chart (probability histogram).** Each value $x$ is represented by a bar whose height equals $P(X = x)$.

**EXEMPLE** **The sum of two dice**

Roll two fair dice and let $X$ be the sum of the two scores. There are 36 equally likely outcomes, and counting those giving each sum yields:

| $x$ | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $36\,P(X{=}x)$ | 1 | 2 | 3 | 4 | 5 | 6 | 5 | 4 | 3 | 2 | 1 |

The famous triangular shape appears clearly on the bar chart below.




Probability histogram of $X$ = sum of two fair dice: a symmetric triangular distribution peaking at $x = 7$.

## 1.6 The two fundamental conditions

Not every list of numbers can be a probability distribution. Since probabilities are numbers between 0 and 1, and since the events $\{X = x_i\}$ partition the whole sample space, we must have:

**PROPERTY** **Fundamental conditions for a probability distribution**

A function $p(x) = P(X = x)$ is the p.m.f. of a discrete random variable if and only if

1. $p(x) \geq 0$ for every value $x$ (in fact $0 \leq p(x) \leq 1$);
2. $\sum_{\text{all } x} p(x) = 1$.

(Not examinable as a proof; it follows directly from the axioms of probability, since the events $\{X = x_i\}$ are mutually exclusive and exhaustive.)

**WARNING** **Probabilities must sum to 1**

The most frequent error in GCE scripts is presenting a "distribution" whose probabilities do not add up to 1 — for example after a rounding error, or after forgetting one value of $X$. Always check the total before moving on; examiners award a mark for this verification.

## 1.7 Finding an unknown constant in a p.m.f.

A classic GCE question gives a p.m.f. containing an unknown constant $k$ and asks you to find it. The only tool needed is the condition $\sum p(x) = 1$.

**METHOD** **Finding an unknown constant in a p.m.f.**

1. Write down every value $x$ that $X$ can take and the expression for $p(x)$ in terms of $k$.
2. Impose the condition $\sum_{x} p(x) = 1$ and solve the resulting equation for $k$.
3. Check that with this value of $k$, every probability satisfies $0 \le p(x) \le 1$.
4. Rewrite the complete distribution table with numerical probabilities.

**WORKED EXAMPLE 1** **Finding the constant $k$**

A discrete random variable $X$ has probability mass function

$$P(X = x) = kx, \qquad x = 1, 2, 3, 4.$$

(a) Find the value of $k$. (b) Write out the probability distribution of $X$ in a table.

**SOLUTION**

(a) The probabilities are $P(X = 1) = k$, $P(X = 2) = 2k$, $P(X = 3) = 3k$, $P(X = 4) = 4k$. Imposing the total probability condition:

$$\sum_{x} P(X = x) = 1 \implies k + 2k + 3k + 4k = 1$$

$$\implies 10k = 1 \implies k = \frac{1}{10}.$$

Check: all probabilities $\frac{1}{10}, \frac{2}{10}, \frac{3}{10}, \frac{4}{10}$ lie in $[0, 1]$. ✓
(b) The distribution is:

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $P(X = x)$ | 0.1 | 0.2 | 0.3 | 0.4 |

Total: $0.1 + 0.2 + 0.3 + 0.4 = 1$. ✓

## 1.8 Computing probabilities of compound events

Once the distribution is known, the probability of any event defined by $X$ is found by **adding** the probabilities of the individual values inside the event (the events $\{X = x\}$ are mutually exclusive). In particular:

**PROPERTY** **Probabilities of compound events**

For a discrete random variable $X$:

$$P(X \geq a) = \sum_{x \geq a} P(X = x), \qquad P(a < X \leq b) = \sum_{a < x \leq b} P(X = x),$$

$$P(X \neq a) = 1 - P(X = a).$$

**WORKED EXAMPLE 2** **Using the distribution of the sum of two dice**

Using the distribution of $X$ = sum of two fair dice, find (a) $P(X \geq 10)$; (b) $P(4 < X \leq 7)$; (c) $P(X \neq 7)$.

**SOLUTION**

From the table, $P(X = x) = \dfrac{6 - |x - 7|}{36}$ for $x = 2, \ldots, 12$.

(a) $P(X \geq 10) = P(X = 10) + P(X = 11) + P(X = 12) = \dfrac{3}{36} + \dfrac{2}{36} + \dfrac{1}{36} = \dfrac{6}{36} = \dfrac{1}{6}$.

(b) The event $4 < X \leq 7$ contains the values $5, 6, 7$:

$$P(4 < X \leq 7) = \frac{4}{36} + \frac{5}{36} + \frac{6}{36} = \frac{15}{36} = \frac{5}{12}.$$

(c) By the complement rule:

$$P(X \neq 7) = 1 - P(X = 7) = 1 - \frac{6}{36} = \frac{5}{6}.$$

**WARNING** **Strict versus non-strict inequalities**

Be careful with the boundary: $P(X \geq 10)$ includes $x = 10$, while $P(X > 10)$ excludes it. For a discrete variable these are different numbers — unlike for continuous variables. Read the inequality sign twice before adding.

**EXERCISE 1** **Practice**

A discrete random variable $X$ takes the values $0, 1, 2, 3$ with

$$P(X = x) = k(x + 1), \qquad x = 0, 1, 2, 3.$$

1. Show that $k = \frac{1}{10}$ and write out the full distribution table.
2. Find $P(X \geq 2)$, $P(0 < X \leq 2)$ and $P(X \neq 1)$.

**SOLUTION** **Exercise 1**

1. $\sum p(x) = k(1+2+3+4) = 10k = 1$, so $k = \frac{1}{10}$. The table is:

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $P(X = x)$ | 0.1 | 0.2 | 0.3 | 0.4 |

All probabilities are in $[0, 1]$ and sum to 1. ✓
2. $P(X \geq 2) = 0.3 + 0.4 = 0.7$; $P(0 < X \leq 2) = P(X = 1) + P(X = 2) = 0.2 + 0.3 = 0.5$; $P(X \neq 1) = 1 - 0.2 = 0.8$.

**KEY POINTS** **Key points of this section**

- A **random variable** is a function $X : \Omega \to \mathbb{R}$ assigning a number to each outcome; it is **discrete** if its values can be listed.
- The **probability distribution** of $X$ is the set of pairs $(x, P(X = x))$; it may be given as a table, a formula, or a bar chart.
- Every distribution satisfies $0 \leq P(X = x) \leq 1$ and $\sum_x P(X = x) = 1$.
- An unknown constant in a p.m.f. is found by solving $\sum p(x) = 1$, then checking non-negativity.
- Compound probabilities are sums: $P(X \geq a) = \sum_{x \geq a} p(x)$, and $P(X \neq a) = 1 - P(X = a)$.

Our Mokolo trader now has her "single table": the list of possible daily demands with their probabilities. But a table alone does not tell her how many cards to stock. For that she needs a summary number — the expected demand — which is the subject of the next section.

# 2 THE PROBABILITY MASS FUNCTION AND EXPECTATION

## 2.1 From distributions to the of a random variable

In Section 1, we defined a of a, and a discrete random variable $X$, as the function$p$defined by $p(x = P(X = x)$, where $x$ is a. The function $p$ is called the probability mass function (p.m.f). We now make this idea precise.

## 2.2 The probability of a random variable

is a random variable, and the p.m.f of a of a of a

# 3 VARIANCE AND STANDARD DEVIATION OF A DISCRETE RANDOM VARIABLE

In the previous section we defined the **expectation** $E(X)$ of a discrete random variable and saw that it plays the role of an average value: it tells us where the distribution is centred. But two distributions can be centred at the very same point and yet behave completely differently. A trader in Bamenda whose daily profit is always close to 5 000 FCFA and a trader whose profit swings wildly between -2 000 and 12 000 FCFA may have the same average profit of 5 000 FCFA. The mean alone cannot see this difference. We need a second number that measures the **spread** of a distribution around its mean. That number is the **variance**, and its square root, the **standard deviation**.

## 3.1 Why the mean is not enough: a motivating example

**EXEMPLE** **Two games, one mean**

Consider two games of chance at a school fair. In Game A you win a random amount $X$; in Game B you win $Y$. Their distributions are:

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $P(X = x)$ | 0.1 | 0.2 | 0.4 | 0.2 | 0.1 |

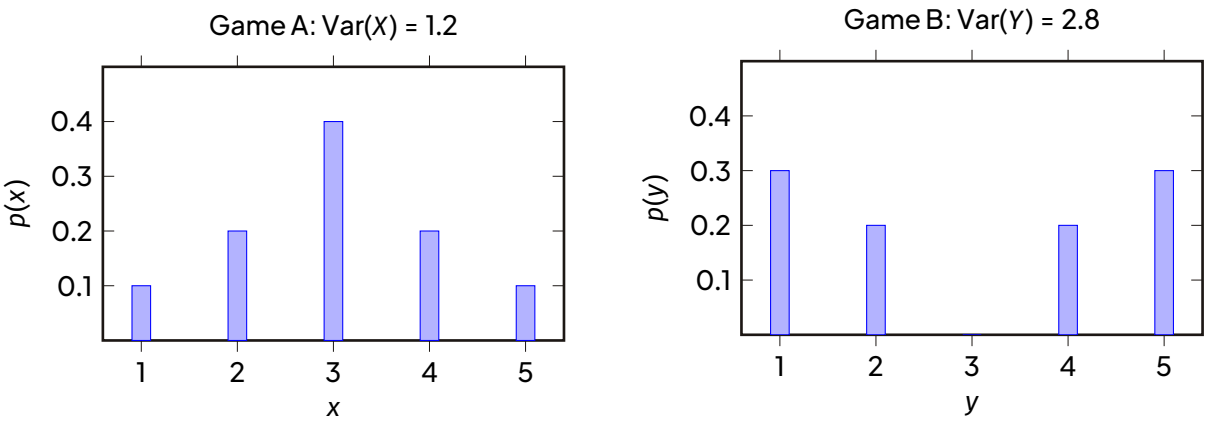
| $y$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $P(Y = y)$ | 0.3 | 0.2 | 0 | 0.2 | 0.3 |

A quick computation gives

$$E(X) = 1(0.1) + 2(0.2) + 3(0.4) + 4(0.2) + 5(0.1) = 3,$$

$$E(Y) = 1(0.3) + 2(0.2) + 3(0) + 4(0.2) + 5(0.3) = 3.$$

Both games have the same mean $\mu = 3$. Yet Game A usually pays close to 3, while Game B frequently pays the extreme values 1 or 5 and never pays 3. If you dislike risk, you prefer Game A. The expectation cannot distinguish them; the variance will.




Two distributions with the same mean $\mu = 3$ but different spreads. Game A is concentrated around the mean (small variance); Game B is spread out (large variance).

## 3.2 Definition of the variance and the standard deviation

How should we measure spread? A natural idea is to look at the deviation of each value from the mean, $x - \mu$. But the average deviation is useless:

$$E(X - \mu) = E(X) - \mu = \mu - \mu = 0,$$

because positive and negative deviations cancel out. To kill the signs we square the deviations (squaring also penalises large deviations more heavily than small ones, which is exactly what we want). This leads to the official definition.

**DEFINITION** **Variance and standard deviation**

Let $X$ be a discrete random variable with mean $\mu = E(X)$. The **variance** of $X$ is

$$\operatorname{Var}(X) = E\big[(X - \mu)^2\big] = \sum_{x} (x - \mu)^2 \, p(x),$$

where the sum runs over all values of $X$ and $p(x) = P(X = x)$. The **standard deviation** of $X$ is

$$\sigma = \sqrt{\operatorname{Var}(X)}.$$

**WARNING** **Units**

The variance is measured in squared units: if $X$ is a profit in FCFA, $\operatorname{Var}(X)$ is in FCFA$^2$, which has no direct interpretation. The standard deviation $\sigma$ is in the same units as $X$ (FCFA), so it is $\sigma$ that we quote when interpreting spread in a real context: "the daily profit is 5 000 FCFA on average, with a standard deviation of 800 FCFA".

## 3.3 The computational formula

Applying the definition directly forces us to compute every deviation $(x - \mu)$, square it, and multiply by $p(x)$: long and error-prone, especially when $\mu$ is not an integer. The following theorem gives a much faster route.

**PROPERTY** **Computational formula for the variance (proof examinable)**

For any discrete random variable $X$,

$$\operatorname{Var}(X) = E(X^2) - [E(X)]^2.$$

Proof. Write $\mu = E(X)$. By definition and linearity of expectation,

$$\begin{aligned}\operatorname{Var}(X) &= E\left[(X - \mu)^2\right] = E\left[X^2 - 2\mu X + \mu^2\right] \\ &= E(X^2) - 2\mu\, E(X) + \mu^2 \\ &= E(X^2) - 2\mu^2 + \mu^2 = E(X^2) - \mu^2 = E(X^2) - [E(X)]^2. \qquad \square\end{aligned}$$

**WARNING** **A classic bracket error**

Never write $\operatorname{Var}(X) = E(X^2) - E(X)^2$ carelessly: the mean is squared, $[E(X)]^2$, not the variable. $E(X^2)$ (mean of the squares) and $[E(X)]^2$ (square of the mean) are different numbers, and their difference is precisely the variance. Note also that since $(X - \mu)^2 \geq 0$, we always have $E(X^2) \geq [E(X)]^2$.

**METHOD** **Computing $\operatorname{Var}(X)$ from a table**

1. Build a table with columns $x$, $p(x)$, $x\,p(x)$ and $x^2\,p(x)$.
2. Sum the third column to get $E(X)$; sum the fourth column to get $E(X^2)$.
3. Check that the $p(x)$ column sums to 1.
4. Apply $\operatorname{Var}(X) = E(X^2) - [E(X)]^2$, then $\sigma = \sqrt{\operatorname{Var}(X)}$.

**WORKED EXAMPLE 3** **Computing $E(X)$, $E(X^2)$, $\operatorname{Var}(X)$ and $\sigma$**

The number $X$ of phones sold per day by a shop in Douala has distribution:

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $p(x)$ | 0.1 | 0.3 | 0.3 | 0.2 | 0.1 |

Find $E(X)$, $E(X^2)$, $\mathrm{Var}(X)$ and the standard deviation $\sigma$.

SOLUTION

Solution. Extend the table:

| $x$ | $p(x)$ | $x\,p(x)$ | $x^2\,p(x)$ |
|---|---|---|---|
| 0 | 0.1 | 0 | 0 |
| 1 | 0.3 | 0.3 | 0.3 |
| 2 | 0.3 | 0.6 | 1.2 |
| 3 | 0.2 | 0.6 | 1.8 |
| 4 | 0.1 | 0.4 | 1.6 |
| Total | 1 | 1.9 | 4.9 |

Hence $E(X) = 1.9$ and $E(X^2) = 4.9$. By the computational formula,

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = 4.9 - (1.9)^2 = 4.9 - 3.61 = 1.29,$$

$$\sigma = \sqrt{1.29} \approx 1.14 \text{ phones.}$$

Interpretation: the shop sells about 1.9 phones per day on average, and the daily sales typically fluctuate by about 1.1 phone around this mean.

## 3.4 Properties of the variance

PROPERTY **Basic properties of the variance**

Let $X$ be a discrete random variable and $a, b \in \mathbb{R}$.

1. $\mathrm{Var}(X) \geq 0$.
2. $\mathrm{Var}(X) = 0$ if and only if $X$ is constant (takes a single value with probability 1).
3. $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$.

Proof of (3). Recall $E(aX + b) = a\mu + b$. Then

$$\mathrm{Var}(aX + b) = E\Big[\big((aX + b) - (a\mu + b)\big)^2\Big] = E\big[a^2(X - \mu)^2\big]$$
$$= a^2 E\big[(X - \mu)^2\big] = a^2\,\mathrm{Var}(X). \qquad \square$$

Property (3) has a very intuitive meaning:

- Adding a constant $b$ shifts the whole distribution without changing its shape: the spread is unchanged, so $b$ disappears from the variance.

- Multiplying by $a$ stretches the axis by $|a|$; since variance involves squared deviations, it is multiplied by $a^2$. Consequently the standard deviation is multiplied by $|a|$: $\sigma_{aX+b} = |a|\,\sigma_X$.

**WARNING** **The two most frequent mistakes**

- $\mathrm{Var}(aX) = a^2\mathrm{Var}(X)$, not $a\,\mathrm{Var}(X)$. For example $\mathrm{Var}(2X) = 4\,\mathrm{Var}(X)$.
- $\mathrm{Var}(X + b) = \mathrm{Var}(X)$, not $\mathrm{Var}(X) + b$. A translation does not change the spread.

Contrast carefully with expectation:

$$E(aX + b) = aE(X) + b \qquad \text{versus} \qquad \mathrm{Var}(aX + b) = a^2\mathrm{Var}(X).$$

**PROPERTY** **Variance of a sum of independent variables (extension, useful for the GCE)**

If $X$ and $Y$ are independent random variables, then

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).$$

The proof uses the fact that for independent variables $E(XY) = E(X)E(Y)$; it is not examinable but the result is extremely useful. Beware: independence is essential — the formula can fail otherwise.

**WORKED EXAMPLE 4** **Finding unknowns given $E(X)$ and $\mathrm{Var}(X)$**

A random variable $X$ takes the values 1, 2 and $k$ with probabilities

$$P(X = 1) = 0.5, \qquad P(X = 2) = 0.3, \qquad P(X = k) = 0.2,$$

where $k > 2$. Given that $E(X) = 2.1$, find $k$, then compute $\mathrm{Var}(X)$ and $\sigma$.

**SOLUTION**

Solution. From the expectation:

$$E(X) = 1(0.5) + 2(0.3) + 0.2k = 1.1 + 0.2k = 2.1 \implies 0.2k = 1 \implies k = 5.$$

Now compute $E(X^2)$:

$$E(X^2) = 1^2(0.5) + 2^2(0.3) + 5^2(0.2) = 0.5 + 1.2 + 5 = 6.7.$$

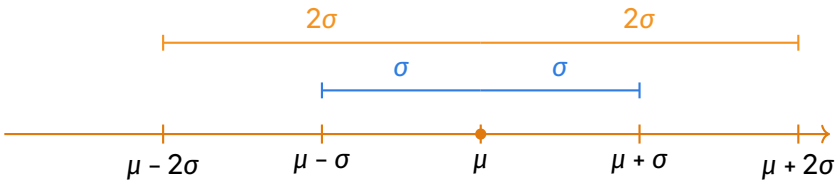
Therefore

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = 6.7 - (2.1)^2 = 6.7 - 4.41 = 2.29,$$

$$\sigma = \sqrt{2.29} \approx 1.51.$$

## 3.5 Interpreting the standard deviation in context

The standard deviation measures a typical distance from the mean. For most distributions met at this level, the great majority of the probability lies within $\mu \pm 2\sigma$.




Number line showing the mean $\mu$ and the bands $\mu \pm \sigma$ and $\mu \pm 2\sigma$. The standard deviation is the natural "ruler" for measuring spread around the mean.

In applications, variance quantifies **risk** and **consistency**:

- Finance and trade. Two investments with the same expected gain are not equivalent: the one with the smaller variance is safer. A trader whose daily sales have a small standard deviation can plan stock confidently.
- Quality control. A machine filling 50 cl bottles in a Douala factory must have a small standard deviation of the filled volume, otherwise many bottles are under- or over-filled.
- Measurements. Repeated measurements of the same quantity give a small variance when the instrument is reliable.

**EXEMPLE** **Using the linearity properties**

Let $X$ be the daily profit (in thousands of FCFA) of a kiosk, with $E(X) = 5$ and $\mathrm{Var}(X) = 4$. A tax of 1 (thousand FCFA) is charged daily, and the remaining profit is doubled by a new sales strategy: $Y = 2(X - 1) = 2X - 2$. Then

$$E(Y) = 2E(X) - 2 = 8, \qquad \mathrm{Var}(Y) = 2^2\,\mathrm{Var}(X) = 16, \qquad \sigma_Y = 4.$$

Note how the shift by −2 changed the mean but left the variance untouched, while doubling quadrupled the variance.

**EXERCISE 2** **Practice**

A discrete random variable $X$ has distribution

| $x$ | −1 | 0 | 1 | 2 |
|---|---|---|---|---|
| $p(x)$ | 0.2 | 0.1 | 0.3 | 0.4 |

1. Compute $E(X)$, $E(X^2)$, $\mathrm{Var}(X)$ and $\sigma$.

2. Let $Y = 3X + 2$. Find $E(Y)$, $\mathrm{Var}(Y)$ and $\sigma_Y$.

**SOLUTION** **Exercise 1**

1. $E(X) = -1(0.2) + 0(0.1) + 1(0.3) + 2(0.4) = 0.9$. $E(X^2) = 1(0.2) + 0(0.1) + 1(0.3) + 4(0.4) = 2.1$. $\mathrm{Var}(X) = 2.1 - (0.9)^2 = 2.1 - 0.81 = 1.29$; $\sigma = \sqrt{1.29} \approx 1.14$.
2. $E(Y) = 3E(X)+2 = 3(0.9)+2 = 4.7$. $\mathrm{Var}(Y) = 3^2\,\mathrm{Var}(X) = 9\times 1.29 = 11.61$; $\sigma_Y = 3\sigma = 3\sqrt{1.29} \approx 3.41$.

**KEY POINTS** **Key points**

- The variance measures the spread of $X$ around its mean: $\mathrm{Var}(X) = E[(X-\mu)^2] = \sum (x-\mu)^2 p(x)$.
- In practice use the computational formula $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$.
- The standard deviation $\sigma = \sqrt{\mathrm{Var}(X)}$ is expressed in the same units as $X$; it is the quantity used for interpretation.
- $\mathrm{Var}(X) \geq 0$, with equality only when $X$ is constant.
- $E(aX + b) = aE(X) + b$ but $\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$: the constant $b$ vanishes and $a$ is squared.
- If $X$ and $Y$ are independent, $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$.

# 4 THE CUMULATIVE DISTRIBUTION FUNCTION, MODE AND MEDIAN

In the previous section we summarised a discrete random variable by two numbers: its expectation $E(X)$, which locates the "centre" of the distribution, and its variance $\mathrm{Var}(X)$, which measures the spread around that centre. But a single number can never tell the whole story. Suppose a student at GBHS Bamenda wants to know: what is the probability that the number of goals scored in a match is at most 2? Or: what is the most likely number of children in a household? Or: what value splits the distribution into two halves? To answer such questions efficiently we need a new tool that accumulates probabilities: the **cumulative distribution function**, and two new summary values: the **mode** and the **median**.

## 4.1 The cumulative distribution function

**Motivation.**

A fair die is rolled and $X$ is the score obtained. The probability mass function is $p(x) = \frac{1}{6}$ for $x =$

1, 2, … , 6. If we ask for $P(X \le 3)$, we must add $p(1) + p(2) + p(3) = \frac{3}{6}$. If we then ask for $P(X \le 4)$, we add one more term. Rather than redoing these sums every time, we compute them once and for all and store the results in a function $F$: this is the cumulative distribution function.

**DEFINITION** **Cumulative distribution function**

Let $X$ be a discrete random variable with probability mass function $p(x) = P(X = x)$. The **cumulative distribution function** (c.d.f) of $X$ is the function $F$ defined for every real number $x$ by

$$F(x) = P(X \le x) = \sum_{t \le x} p(t).$$

**WARNING** **The c.d.f is defined for ALL real $x$**

A very common mistake is to define $F$ only at the values taken by $X$. The c.d.f is defined on the whole of $\mathbb{R}$: for example, with the die above, $F(3.7) = P(X \le 3.7) = p(1) + p(2) + p(3) = \frac{1}{2}$, and $F(-2) = 0$, $F(100) = 1$. Between two consecutive values of $X$, $F$ is constant.

**EXEMPLE** **Constructing $F$ step by step**

Let $X$ have p.m.f given by the table

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $p(x)$ | 0.1 | 0.3 | 0.4 | 0.2 |

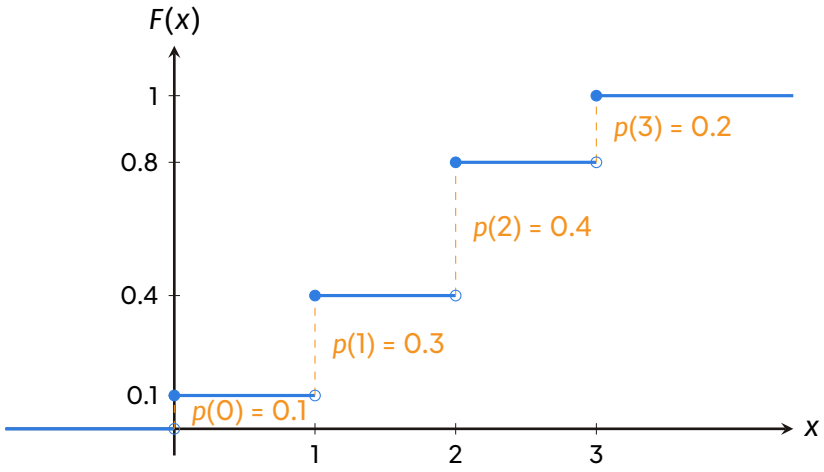
Accumulating the probabilities from left to right:

$$F(0) = 0.1, \quad F(1) = 0.1 + 0.3 = 0.4, \quad F(2) = 0.4 + 0.4 = 0.8, \quad F(3) = 0.8 + 0.2 = 1.$$

Since $F$ is defined for all real $x$, we write it piecewise:

$$F(x) = \begin{cases} 0 & \text{if } x < 0, \\ 0.1 & \text{if } 0 \le x < 1, \\ 0.4 & \text{if } 1 \le x < 2, \\ 0.8 & \text{if } 2 \le x < 3, \\ 1 & \text{if } x \ge 3. \end{cases}$$

Note the shape: $F$ is a step function, constant on each interval, jumping at each value taken by $X$.




The c.d.f $F(x)$ is a step function: closed dots show the value taken at each jump (right-continuity); each jump height equals the p.m.f value $p(x)$.

**PROPERTY** **Properties of the c.d.f**

For any discrete random variable $X$ with c.d.f $F$:

1. $0 \le F(x) \le 1$ for every real $x$;
2. $F$ is **non-decreasing**: if $a \le b$ then $F(a) \le F(b)$;
3. $F(x) \to 0$ as $x \to -\infty$ and $F(x) \to 1$ as $x \to +\infty$; in fact $F(x) = 0$ below the smallest value of $X$ and $F(x) = 1$ from the largest value of $X$ onwards;
4. $F$ is **right-continuous**: at each jump point $a$, $F(a)$ equals the value after the jump (the closed dot is on the upper step).

These properties are not examinable as formal proofs, but you must be able to verify them on a given function.

**PROPERTY** **Computing probabilities with $F$**

For real numbers $a < b$:

$$P(a < X \le b) = F(b) - F(a), \qquad P(X > a) = 1 - F(a), \qquad P(X = a) = F(a) - F(a^-),$$

where $F(a^-)$ denotes the limit of $F$ as $x$ approaches $a$ from the left (the value just before the jump).

**Justification.** The event $\{X \le b\}$ is the disjoint union of $\{X \le a\}$ and $\{a < X \le b\}$, so $F(b) = F(a) + P(a < X \le b)$, which gives the first formula. The second is the complementary event of $\{X \le a\}$. The third says that the probability of a single value is exactly the height of the jump of $F$ at that point.

## 4.2 Recovering the p.m.f from the c.d.f

This is a classic GCE question: the c.d.f is given piecewise and you are asked to find the probability distribution of $X$.

**METHOD** **From the c.d.f to the p.m.f**

1. Locate the **jump points** of $F$: these are the values where the formula changes; they are the values taken by $X$.
2. At each jump point $x$, compute the jump height $p(x) = F(x) - F(x^-)$, i.e. the value of $F$ on that step minus the value on the previous step.
3. Present the p.m.f in a table and check that the probabilities sum to 1.

**WORKED EXAMPLE 5** **A complete GCE-style problem**

The c.d.f of a discrete random variable $X$ is

$$F(x) = \begin{cases} 0 & x < 1, \\ 0.2 & 1 \le x < 2, \\ 0.5 & 2 \le x < 4, \\ 0.9 & 4 \le x < 5, \\ 1 & x \ge 5. \end{cases}$$

Find (a) the p.m.f of $X$; (b) $P(2 < X \le 4)$; (c) $E(X)$ and $\mathrm{Var}(X)$.

**SOLUTION**

(a) The jump points are $x = 1, 2, 4, 5$. The jump heights are:

$$p(1) = 0.2 - 0 = 0.2, \qquad p(2) = 0.5 - 0.2 = 0.3,$$
$$p(4) = 0.9 - 0.5 = 0.4, \qquad p(5) = 1 - 0.9 = 0.1.$$

Check: $0.2 + 0.3 + 0.4 + 0.1 = 1$. ✓
(b) $P(2 < X \le 4) = F(4) - F(2) = 0.9 - 0.5 = 0.4$. (Indeed only $X = 4$ lies in $(2, 4]$.)
(c) Expectation:

$$E(X) = 1(0.2) + 2(0.3) + 4(0.4) + 5(0.1) = 0.2 + 0.6 + 1.6 + 0.5 = 2.9.$$

Second moment:

$$E(X^2) = 1(0.2) + 4(0.3) + 16(0.4) + 25(0.1) = 0.2 + 1.2 + 6.4 + 2.5 = 10.3.$$

Hence

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = 10.3 - 2.9^2 = 10.3 - 8.41 = 1.89.$$

**WARNING** **Exam tip: check $F$ before using it**

When $F$ is given piecewise, always check first that the final piece equals 1, that the first piece equals 0, and that $F$ is non-decreasing. If one of these fails, either the question contains an unknown constant to determine first, or there is an error. For example, if the third piece were $0.9 + k$ instead of 0.9, you would use the condition "last piece = 1" together with the jump structure to find $k$.

## 4.3 The mode of a discrete distribution

**Motivation.**

A trader at Mokolo market records the number $X$ of bags of garri sold per day. For her stock decisions, the most useful number is not the mean but the most frequent daily sale: the value of $X$ that occurs with the greatest probability.

**DEFINITION** **Mode**

The **mode** of a discrete random variable $X$ is the value of $x$ with the greatest probability:

$$\text{mode} = \arg\max_{x} p(x).$$

A distribution with one mode is called **unimodal**; if two values share the same maximal probability, the distribution is **bimodal** (and both values are modes).

**EXEMPLE** **Unimodal and bimodal distributions**

For the distribution $p(0) = 0.1$, $p(1) = 0.3$, $p(2) = 0.4$, $p(3) = 0.2$, the largest probability is $p(2) = 0.4$, so the mode is 2 (unimodal).
For the distribution $p(1) = 0.3$, $p(2) = 0.2$, $p(3) = 0.3$, $p(4) = 0.2$, both $p(1)$ and $p(3)$ equal 0.3: the distribution is bimodal with modes 1 and 3.

## 4.4 The median of a discrete distribution

**Motivation.**

The mean can be dragged away from the "typical" value by a few extreme outcomes. The **median** is a centre that resists this: it is a value $m$ such that at least half the probability lies at or below $m$, and at least half lies at or above $m$.

**DEFINITION** **Median**

A **median** of a discrete random variable $X$ is any value $m$ such that

$$P(X \le m) \ge \tfrac{1}{2} \quad \text{and} \quad P(X \ge m) \ge \tfrac{1}{2}.$$

In practice, the median is read from the c.d.f: it is the smallest value $m$ of $X$ such that

$$F(m) \ge \tfrac{1}{2}.$$

**EXEMPLE** **Reading the median from $F$**

For the distribution $p(0) = 0.1$, $p(1) = 0.3$, $p(2) = 0.4$, $p(3) = 0.2$, the cumulative probabilities are $F(0) = 0.1$, $F(1) = 0.4$, $F(2) = 0.8$, $F(3) = 1$. The smallest $x$ with $F(x) \ge 0.5$ is $x = 2$ (since $F(1) = 0.4 < 0.5$ but $F(2) = 0.8 \ge 0.5$). Hence the median is $m = 2$. Check the definition: $P(X \le 2) = 0.8 \ge 0.5$ and $P(X \ge 2) = 0.4 + 0.2 = 0.6 \ge 0.5$. ✓

**WARNING** **What the median is NOT**

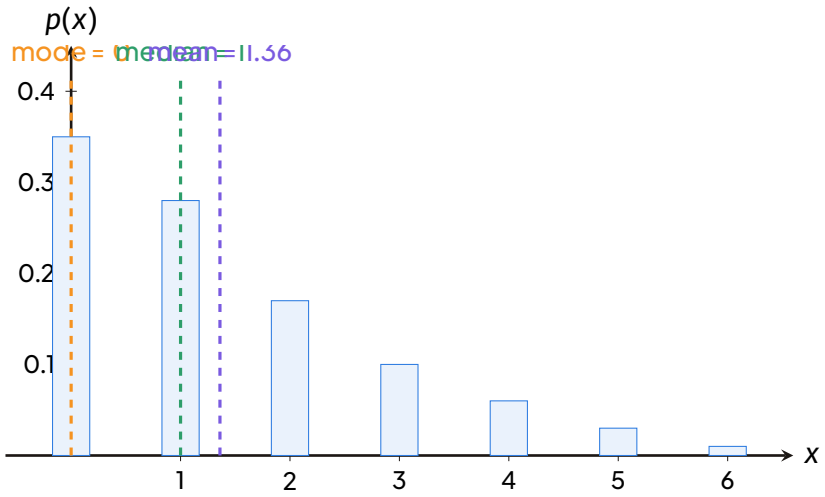
The median is not necessarily $\dfrac{\text{smallest} + \text{largest}}{2}$, and it is not necessarily equal to the mean. It must be read from the cumulative probabilities. Also beware: if $F$ takes the value exactly $0.5$ on a whole step, say $F(2) = 0.5$ and $F(3) = 0.7$, then every real number in $[2, 3)$ satisfies the two inequalities and is a median; by convention at GCE level we quote the smallest value of $X$ with $F(m) \ge 0.5$, here $m = 2$.

## 4.5 Comparing mean, median and mode

For a perfectly symmetric distribution, the mean, median and mode coincide. For a **skewed** distribution they separate in a characteristic order:

- if the distribution has a long tail to the right (positively skewed), then typically mode < median < mean;
- if the tail is to the left (negatively skewed), the order is reversed.

The mean is pulled towards the tail because large values, even rare ones, contribute heavily to $\sum x\,p(x)$. The median and mode are robust: they ignore the size of extreme values.




A right-skewed p.m.f with $p(0) = 0.35$, $p(1) = 0.28$, $p(2) = 0.17$, $p(3) = 0.10$, $p(4) = 0.06$, $p(5) = 0.03$, $p(6) = 0.01$: mode < median < mean. The mean is pulled towards the tail.

**Choosing the right summary in context.**

- Use the **mode** when the question is "which outcome is most likely?" (stock planning, most common number of children per family).
- Use the **median** when you want a typical value unaffected by rare extremes (incomes, waiting times).
- Use the **mean** when totals matter: the mean multiplied by the number of trials gives the expected total (expected total revenue, expected number of successes in $n$ games).

**WORKED EXAMPLE 6** **Mode, median and mean from a c.d.f**

The c.d.f of the number $X$ of customers served per hour in a cybercafé in Yaoundé is

$$F(x) = \begin{cases} 0 & x < 0, \\ 0.15 & 0 \le x < 1, \\ 0.45 & 1 \le x < 2, \\ 0.75 & 2 \le x < 3, \\ 0.92 & 3 \le x < 4, \\ 1 & x \ge 4. \end{cases}$$

Find the p.m.f, the mode, the median and the mean, and comment on the skewness.

**SOLUTION**

p.m.f (jump heights): $p(0) = 0.15$, $p(1) = 0.45 - 0.15 = 0.30$, $p(2) = 0.75 - 0.45 = 0.30$, $p(3) = 0.92 - 0.75 = 0.17$, $p(4) = 1 - 0.92 = 0.08$. Sum: 1. ✓

Mode: $p(1) = p(2) = 0.30$ are both maximal, so the distribution is bimodal with modes 1 and 2.
Median: $F(1) = 0.45 < 0.5$ and $F(2) = 0.75 \geq 0.5$, so the median is $m = 2$.
Mean:

$$E(X) = 0(0.15) + 1(0.30) + 2(0.30) + 3(0.17) + 4(0.08) = 0.30 + 0.60 + 0.51 + 0.32 = 1.73.$$

Comment: the mean (1.73) is slightly below the median (2): the distribution is mildly skewed to the left, the small values pulling the mean down.

**EXERCISE 3** **Consolidation**

A discrete random variable $X$ has c.d.f

$$F(x) = \begin{cases} 0 & x < 1, \\ 0.1 & 1 \leq x < 3, \\ 0.4 & 3 \leq x < 6, \\ 0.85 & 6 \leq x < 8, \\ 1 & x \geq 8. \end{cases}$$

1. Write down the p.m.f of $X$ in a table.
2. Compute $P(3 < X \leq 6)$ and $P(X > 3)$.
3. Find the mode and the median of $X$.
4. Compute $E(X)$ and $\mathrm{Var}(X)$.

**SOLUTION** **Exercise 1**

1. Jump heights: $p(1) = 0.1$, $p(3) = 0.4-0.1 = 0.3$, $p(6) = 0.85-0.4 = 0.45$, $p(8) = 1-0.85 = 0.15$. Sum = 1. ✓
2. $P(3 < X \leq 6) = F(6) - F(3) = 0.85 - 0.4 = 0.45$; $P(X > 3) = 1 - F(3) = 1 - 0.4 = 0.6$.
3. Mode: largest probability is $p(6) = 0.45$, so the mode is 6. Median: $F(3) = 0.4 < 0.5$ and $F(6) = 0.85 \geq 0.5$, so the median is 6.
4. $E(X) = 1(0.1) + 3(0.3) + 6(0.45) + 8(0.15) = 0.1 + 0.9 + 2.7 + 1.2 = 4.9$. $E(X^2) = 1(0.1) + 9(0.3) + 36(0.45) + 64(0.15) = 0.1 + 2.7 + 16.2 + 9.6 = 28.6$. $\mathrm{Var}(X) = 28.6 - 4.9^2 = 28.6 - 24.01 = 4.59$.

**KEY POINTS** **Key points of this section**

- The c.d.f is $F(x) = P(X \leq x) = \sum_{t \leq x} p(t)$, defined for every real $x$; it is a non-decreasing, right-continuous step function with $F \to 0$ at $-\infty$ and $F \to 1$ at $+\infty$.
- $P(a < X \leq b) = F(b) - F(a)$ and $P(X > a) = 1 - F(a)$.
- The p.m.f is recovered from the jumps: $p(x) = F(x) - F(x^-)$.
- The **mode** is the value with the greatest probability (a distribution may be bimodal).
- The **median** is the smallest value $m$ with $F(m) \geq \frac{1}{2}$; it is read from cumulative probabilities, never computed as an average of extreme values.
- For right-skewed distributions: mode < median < mean; choose the summary according to the context.

DID YOU KNOW? **Why medians matter in real life**

When newspapers report "average income", they often quote the median rather than the mean: a few very high salaries would pull the mean far above what a typical worker earns, while the median stays at the level of the person in the middle. The same robustness makes the median essential in medicine (survival times) and in queueing studies (waiting times at a hospital or a mobile-money kiosk). Knowing how to read it from a cumulative distribution is a genuinely useful life skill, not just an exam technique!

# 5 MOMENT GENERATING FUNCTIONS

In the previous section we met the cumulative distribution function, the mode and the median — tools that describe a distribution globally. We now introduce a single function that encodes the whole distribution in one formula: the **moment generating function** (m.g.f). Its power is this: by simply differentiating one function and substituting $t = 0$, we recover the expectation, the second moment, and hence the variance — no long sums of $x\,p(x)$ or $x^2 p(x)$ needed. This is a favourite GCE Advanced Level topic because it rewards careful algebra and clean differentiation.

## 5.1 Moments of a distribution

EXPERIMENT **Why "moments"?**

Think of the marks of a class in a Further Mathematics mock examination. The mean tells you the "centre" of the marks. But two classes can have the same mean and very different spreads: one class where everyone scored near 12/20, another with marks scattered

from 3 to 19. To describe the spread we need information about $X^2$ as well as $X$. More generally, the quantities $E(X)$, $E(X^2)$, $E(X^3)$, … capture progressively finer features of the distribution. They are called the **moments** of $X$, by analogy with the moments of masses in mechanics.

**DEFINITION** **Moment of order $r$**

Let $X$ be a discrete random variable. The **moment of order $r$** (or $r$-th moment about the origin) of $X$ is

$$\mu_r' = E(X^r) = \sum_x x^r \, P(X = x),$$

provided the sum exists. In particular:

- the first moment is the expectation: $\mu_1' = E(X)$;
- the second moment is $\mu_2' = E(X^2)$, which appears in the variance formula $\operatorname{Var}(X) = E(X^2) - [E(X)]^2 = \mu_2' - (\mu_1')^2$.

So computing expectations and variances is really the task of computing moments. The moment generating function is a machine that produces all the moments at once.

## 5.2 Definition of the moment generating function

**DEFINITION** **Moment generating function**

Let $X$ be a discrete random variable with probability mass function $p(x) = P(X = x)$. The **moment generating function** of $X$ is

$$M_X(t) = E\left(e^{tX}\right) = \sum_x e^{tx} \, p(x),$$

defined for all real values of $t$ for which this sum exists (converges). The set of such $t$ always contains an interval around 0.

**WARNING** **It is $e^{tx}$, not $e^t$**

The expression $e^{tX}$ means the exponential of the product $tX$. A very common error is to compute $\sum_x e^t \, p(x)$ instead of $\sum_x e^{tx} \, p(x)$. The exponent must contain the value $x$ of the random variable: each value $x$ contributes $e^{tx} \, p(x)$. Since $\sum_x e^t p(x) = e^t \sum_x p(x) = e^t$, that mistake destroys all the information about the distribution.

**PROPERTY** **Value at $t = 0$**

For every random variable $X$,

$$M_X(0) = 1.$$

**Proof (examinable, one line).** Substituting $t = 0$: $M_X(0) = \sum_x e^0\, p(x) = \sum_x p(x) = 1$, since the probabilities of a distribution sum to 1. ■

This gives an instant sanity check: whatever m.g.f you compute, setting $t = 0$ must give 1.

## 5.3 Computing the m.g.f of simple distributions

**EXEMPLE** **Single-point distribution**

Let $X$ take the single value $c$ with probability 1. Then

$$M_X(t) = e^{tc} \cdot 1 = e^{tc}.$$

Check: $M_X(0) = e^0 = 1$. ✓

**EXEMPLE** **Two-point (Bernoulli-type) distribution**

Let $X$ take the value 1 with probability $p$ and 0 with probability $q = 1 - p$. Then

$$M_X(t) = e^{t\cdot 0} q + e^{t\cdot 1} p = q + pe^t.$$

Check: $M_X(0) = q + p = 1$. ✓

**EXEMPLE** **Uniform distribution on $\{1, 2, \dots, n\}$**

Let $P(X = k) = \dfrac{1}{n}$ for $k = 1, 2, \dots, n$. Then

$$M_X(t) = \frac{1}{n}\sum_{k=1}^{n} e^{tk} = \frac{1}{n}\left(e^t + e^{2t} + \cdots + e^{nt}\right).$$

For $t \neq 0$ this is a geometric series with ratio $e^t$, so

$$M_X(t) = \frac{e^t\left(e^{nt} - 1\right)}{n\left(e^t - 1\right)}.$$

At $t = 0$ the formula is understood as the limit $M_X(0) = 1$ (or directly from the definition).

**WORKED EXAMPLE 7** **M.g.f of a geometric-type distribution**

A random variable $X$ has p.m.f

$$P(X = k) = \left(\frac{1}{2}\right)^k, \qquad k = 1, 2, 3, \ldots$$

Find the moment generating function of $X$ and state the values of $t$ for which it exists.

**SOLUTION**

Solution. By definition,

$$M_X(t) = \sum_{k=1}^{\infty} e^{tk}\left(\frac{1}{2}\right)^k = \sum_{k=1}^{\infty}\left(\frac{e^t}{2}\right)^k.$$

This is a geometric series with first term $\frac{e^t}{2}$ and common ratio $r = \frac{e^t}{2}$. It converges if and only if $|r| < 1$, i.e. $e^t < 2$, i.e. $t < \ln 2$. Summing the series:

$$M_X(t) = \frac{\frac{e^t}{2}}{1 - \frac{e^t}{2}} = \frac{e^t}{2 - e^t}, \qquad t < \ln 2.$$

Check: $M_X(0) = \frac{1}{2-1} = 1$. ✓

## 5.4 Why it works: the series expansion

The secret of the m.g.f is the power series of the exponential. Recall that for every real $u$,

$$e^u = 1 + u + \frac{u^2}{2!} + \frac{u^3}{3!} + \cdots = \sum_{r=0}^{\infty} \frac{u^r}{r!}.$$

Applying this with $u = tX$ and taking expectations term by term (justified because the series converges absolutely for $t$ in an interval around 0):

$$M_X(t) = E\left(e^{tX}\right) = E\left(\sum_{r=0}^{\infty} \frac{t^r X^r}{r!}\right) = \sum_{r=0}^{\infty} \frac{t^r}{r!} E(X^r).$$

**KEY POINTS** **The m.g.f as a storehouse of moments**

$$M_X(t) = \sum_{r=0}^{\infty} \frac{t^r}{r!} E(X^r) = 1 + t\,E(X) + \frac{t^2}{2!} E(X^2) + \frac{t^3}{3!} E(X^3) + \cdots$$

The moment $E(X^r)$ is the coefficient of $\frac{t^r}{r!}$ in the expansion of $M_X(t)$. The m.g.f "generates" the moments — hence its name.

## 5.5 Extracting moments by differentiation

Differentiate the expansion with respect to $t$:

$$M_X'(t) = E(X) + t\,E(X^2) + \frac{t^2}{2!}E(X^3) + \cdots$$

Setting $t = 0$ kills every term except the first: $M_X'(0) = E(X)$. Differentiating again,

$$M_X''(t) = E(X^2) + t\,E(X^3) + \cdots \quad \Longrightarrow \quad M_X''(0) = E(X^2).$$

Continuing in this way gives the general result.

**PROPERTY** **Moments from the m.g.f**

If the m.g.f $M_X(t)$ exists in an interval around $t = 0$, then for every positive integer $r$,

$$E(X^r) = M_X^{(r)}(0),$$

where $M_X^{(r)}$ denotes the $r$-th derivative of $M_X$ with respect to $t$. In particular,

$$E(X) = M_X'(0), \qquad E(X^2) = M_X''(0), \qquad \mathrm{Var}(X) = M_X''(0) - \left[M_X'(0)\right]^2.$$

(The derivation above, via the series expansion, is examinable.)

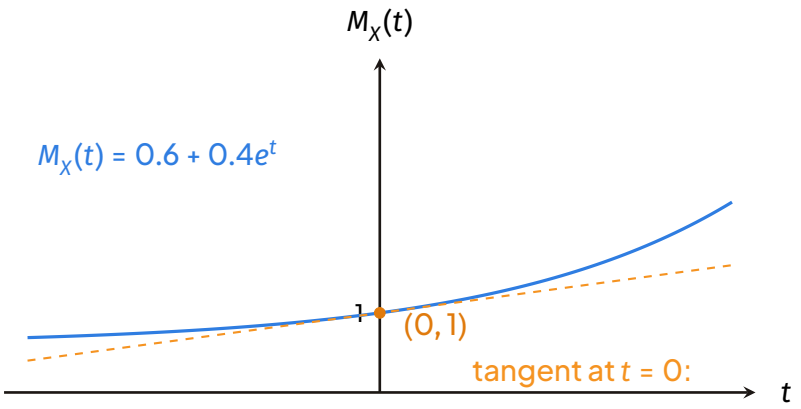
**METHOD** **Finding moments from an m.g.f**

1. Write down $M_X(t)$ and check that $M_X(0) = 1$.
2. Differentiate with respect to $t$ to obtain $M_X'(t)$, then $M_X''(t)$, differentiating as many times as the moment required.
3. Only now substitute $t = 0$: read off $E(X) = M_X'(0)$, $E(X^2) = M_X''(0)$.
4. Compute $\mathrm{Var}(X) = M_X''(0) - [M_X'(0)]^2$.

Never substitute $t = 0$ before differentiating: $M_X(0) = 1$ is a constant, and the derivative of a constant is 0 — you would wrongly conclude that every expectation is zero!

**WARNING** **Differentiate first, substitute after**

A classic GCE error: candidates compute $M_X(0) = 1$, then "differentiate" to get 0 and write $E(X) = 0$. The substitution $t = 0$ must come after all differentiations. Also remember the quotient/product rules when differentiating expressions like $\dfrac{e^t}{2 - e^t}$.




The m.g.f $M_X(t) = 0.6 + 0.4e^t$ of the two-point distribution $P(X = 0) = 0.6$, $P(X = 1) = 0.4$. Every m.g.f passes through $(0, 1)$; the slope of the tangent there is $M_X'(0) = E(X)$.

## 5.6 Two powerful properties

**PROPERTY** **M.g.f of a sum of independent variables**

If $X$ and $Y$ are independent random variables, then

$$M_{X+Y}(t) = M_X(t)\,M_Y(t).$$

Indeed, $M_{X+Y}(t) = E(e^{t(X+Y)}) = E(e^{tX}e^{tY}) = E(e^{tX})\,E(e^{tY})$ by independence. Moreover (stated without proof): the m.g.f determines the distribution uniquely — two random variables with the same m.g.f (on an interval around 0) have the same distribution. This is why recognising a computed m.g.f as that of a known distribution allows you to identify the distribution immediately.

## 5.7 Synthesis worked example

**WORKED EXAMPLE 8** **From p.m.f to m.g.f to moments — full synthesis**

A discrete random variable $X$ has probability mass function

$$P(X = 0) = \tfrac{1}{4}, \qquad P(X = 1) = \tfrac{1}{2}, \qquad P(X = 2) = \tfrac{1}{4}.$$

1. Find the moment generating function $M_X(t)$.
2. Use $M_X$ to find $E(X)$ and $\mathrm{Var}(X)$.
3. Cross-check your answers directly from the p.m.f.

**SOLUTION**

Solution.

1. The m.g.f. By definition,

$$M_X(t) = \sum_x e^{tx}p(x) = \tfrac{1}{4}e^0 + \tfrac{1}{2}e^t + \tfrac{1}{4}e^{2t} = \frac{1}{4} + \frac{1}{2}e^t + \frac{1}{4}e^{2t}.$$

Check: $M_X(0) = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$. ✓ (One may also factorise: $M_X(t) = \frac{1}{4}(1 + e^t)^2$ — note this is $\left(\frac{1}{2} + \frac{1}{2}e^t\right)^2$, the square of a Bernoulli m.g.f, confirming that $X$ is the sum of two independent Bernoulli$\left(\frac{1}{2}\right)$ variables.)

2. Moments by differentiation. Differentiate with respect to $t$:

$$M_X'(t) = \frac{1}{2}e^t + \frac{1}{2}e^{2t}, \qquad M_X''(t) = \frac{1}{2}e^t + e^{2t}.$$

Now substitute $t = 0$:

$$E(X) = M_X'(0) = \frac{1}{2} + \frac{1}{2} = 1, \qquad E(X^2) = M_X''(0) = \frac{1}{2} + 1 = \frac{3}{2}.$$

Hence

$$\mathrm{Var}(X) = M_X''(0) - [M_X'(0)]^2 = \frac{3}{2} - 1^2 = \frac{1}{2}.$$

3. Direct cross-check from the p.m.f.

$$E(X) = 0 \cdot \tfrac{1}{4} + 1 \cdot \tfrac{1}{2} + 2 \cdot \tfrac{1}{4} = 1, \qquad E(X^2) = 0 \cdot \tfrac{1}{4} + 1 \cdot \tfrac{1}{2} + 4 \cdot \tfrac{1}{4} = \tfrac{3}{2},$$

so $\mathrm{Var}(X) = \frac{3}{2} - 1 = \frac{1}{2}$. Both methods agree. ✓

**WARNING** **Exam tip: always cross-check**

Whenever a GCE question asks you to find $E(X)$ and $\mathrm{Var}(X)$ via the m.g.f, quickly recompute $E(X)$ and $E(X^2)$ directly from the p.m.f on the side of your script. If $M_X'(0) \neq \sum x\,p(x)$, you have made a differentiation or algebra error — this simple check catches most sign and coefficient mistakes before they cost you marks.

**EXERCISE 4** **Practice**

A random variable $X$ has p.m.f $P(X = 1) = \frac{1}{3}$, $P(X = 2) = \frac{2}{3}$.

1. Show that $M_X(t) = \dfrac{e^t + 2e^{2t}}{3}$ and verify that $M_X(0) = 1$.
2. Find $E(X)$ and $\mathrm{Var}(X)$ using the m.g.f.

3. Confirm your results directly from the p.m.f.

**SOLUTION** **Exercise 1**

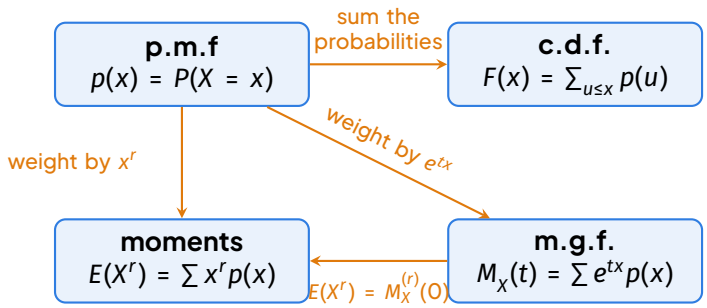
1. $M_X(t) = \frac{1}{3}e^t + \frac{2}{3}e^{2t} = \frac{e^t + 2e^{2t}}{3}$; $M_X(0) = \frac{1+2}{3} = 1$. ✓

2. $M_X'(t) = \frac{e^t + 4e^{2t}}{3}$, so $E(X) = M_X'(0) = \frac{1+4}{3} = \frac{5}{3}$. $M_X''(t) = \frac{e^t + 8e^{2t}}{3}$, so $E(X^2) = M_X''(0) = \frac{9}{3} = 3$.
Hence $\mathrm{Var}(X) = 3 - \left(\frac{5}{3}\right)^2 = 3 - \frac{25}{9} = \frac{2}{9}$.

3. Directly: $E(X) = \frac{1}{3}(1) + \frac{2}{3}(2) = \frac{5}{3}$; $E(X^2) = \frac{1}{3}(1) + \frac{2}{3}(4) = 3$; $\mathrm{Var}(X) = 3 - \frac{25}{9} = \frac{2}{9}$. Agreement confirmed. ✓

## 5.8 Chapter synthesis and GCE revision

The whole chapter fits into one picture: the p.m.f is the fundamental object; everything else is derived from it.




Concept map of the chapter: the p.m.f generates the c.d.f., the moments and the m.g.f.; the m.g.f hands the moments back by differentiation.

| Tool | Definition | Use |
|---|---|---|
| p.m.f. | $p(x) = P(X = x)$, with $\sum p(x) = 1$ | Defines the distribution completely. |
| Expectation | $E(X) = \sum x\,p(x)$ | Average value; first moment. |
| Variance | $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$ | Measures spread about the mean. |
| c.d.f. | $F(x) = P(X \le x)$, a step function | Gives cumulative probabilities; $P(a < X \le b) = F(b) - F(a)$. |
| Mode | Value(s) of $x$ with largest $p(x)$ | Most likely value. |
| Median | Any $m$ with $P(X \le m) \ge \frac{1}{2}$ and $P(X \ge m) \ge \frac{1}{2}$ | Middle value; read from the c.d.f. |
| m.g.f. | $M_X(t) = \sum e^{tx}p(x)$ | Generates all moments: $E(X^r) = M_X^{(r)}(0)$; identifies distributions. |

**KEY POINTS** **GCE exam techniques for this chapter**

- Always check $\sum p(x) = 1$ first — it often determines an unknown constant in the p.m.f.
- Use $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$; never confuse $E(X^2)$ with $[E(X)]^2$.
- For the c.d.f., remember it is a step function, constant between consecutive values of $X$, right-continuous at jumps.
- For the m.g.f.: differentiate first, substitute $t = 0$ afterwards; check $M_X(0) = 1$; cross-check $M_X'(0)$ and $M_X''(0)$ against direct computation.
- If you recognise an m.g.f as that of a known distribution, state the distribution — uniqueness guarantees you are right.

**DID YOU KNOW?** **Why "generating" functions matter**

Generating functions are one of the great unifying ideas of mathematics: a whole sequence of numbers (here the moments $E(X), E(X^2), E(X^3), \ldots$) is packed into a single function whose algebra mirrors the structure of the sequence. The same idea powers the study of the binomial and Poisson distributions you will meet in statistics, and even appears in combinatorics and number theory. Mastering $M_X(t)$ now will make those later chapters feel remarkably easy.

# 6 INTEGRATION ACTIVITY

# 7 INTEGRATION ACTIVITY: THE MOBILE MONEY AGENT'S DILEMMA

## 7.1 Aims of the activity

This integration activity closes the chapter by bringing all the notions studied together in a single, authentic decision-making problem. By the end of the activity, you should be able to:

- construct a **probability mass function** (p.m.f) from observed frequencies and verify that it defines a valid probability distribution;
- compute the **expectation** $E(X)$, $E(X^2)$, the **variance** $\mathrm{Var}(X)$ and the **standard deviation** $\sigma(X)$, and interpret each of them in a concrete business context;
- build and sketch the **cumulative distribution function** (c.d.f) $F(x)$, and read from it the **mode** and the **median** of the distribution;
- use the **linearity of expectation** and the rule $\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$ to analyse a profit function of the form $Y = aX + b$;
- (extension) write down the **moment generating function** $M_X(t)$ and re-derive $E(X)$ and $\mathrm{Var}(X)$ from $M_X'(0)$ and $M_X''(0)$;
- communicate a justified, quantitative piece of advice to a decision-maker, as expected in the GCE Advanced Level examination.

KEY POINTS **Skills mobilised**

p.m.f → validity check → $E(X)$, $E(X^2)$, $\mathrm{Var}(X)$, $\sigma$ → c.d.f table and step graph → mode and median → expectation and variance of $aX + b$ → m.g.f and its derivatives at $t = 0$.

## 7.2 Situation

**The Mobile Money Agent's Dilemma.**

Madame Etongwe runs a small mobile money kiosk at Molyko Junction in Buea. Her main source of income is the commission paid on cash withdrawals: for each withdrawal transaction she processes, the operator pays her a commission of 200 FCFA. However, simply keeping the kiosk open (rent contribution, electricity for the point-of-sale terminal, float management, security) costs her 500 FCFA per hour, whether customers come or not.

Business is irregular: some hours are quiet, others are busy. To understand her situation, Madame Etongwe records, over a large number of working hours spread across several weeks, the number $X$ of cash-withdrawal customers served per hour. She summarises her observations as rel-

ative frequencies, which — because the record covers a long period — she decides to treat as probabilities:

| Number of withdrawals $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Relative frequency (probability) | 0.10 | 0.20 | 0.30 | 0.25 | 0.15 |

Madame Etongwe asks herself two questions:

1. "Is my kiosk viable? On average, do I earn more than it costs me to stay open?"
2. "How risky is the business? Are my hourly earnings stable or wildly unpredictable?"

Working in groups, you will answer her questions rigorously, using the full toolkit of this chapter: the p.m.f, expectation, variance, the c.d.f, the mode, the median, and the moment generating function. Each group will then present its advice to the class, justifying every numerical claim.

## 7.3 Tasks

**Part A — Building the probability distribution**

1. Let $X$ be the number of cash-withdrawal customers served in a randomly chosen hour. Write down the p.m.f of $X$ in a table, giving $P(X = x)$ for each possible value of $x$.
2. Verify that this table defines a valid probability distribution, stating precisely the two conditions you check.
3. Represent the p.m.f by a bar chart (probability on the vertical axis, $x$ on the horizontal axis).

**Part B — Expectation and variance of $X$**

4. Compute $E(X)$. Interpret the result in one sentence that Madame Etongwe can understand.
5. Compute $E(X^2)$, then deduce $\mathrm{Var}(X)$ using the formula $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$.
6. Compute the standard deviation $\sigma(X)$, correct to 2 decimal places. What does it tell the agent about the variability of her hourly customer flow?

**Part C — The c.d.f, the mode and the median**

7. Construct the cumulative distribution function $F(x) = P(X \leq x)$ as a table for $x = 0, 1, 2, 3, 4$, and write $F(x)$ as a piecewise function valid for all real $x$.
8. Sketch the graph of $F$ (a step function), marking clearly the closed and open ends of each step.
9. Read from the p.m.f the **mode** of $X$, and from the c.d.f the **median** of $X$, justifying each answer with the definitions.

**Part D — The profit analysis**

10. The hourly profit (in FCFA) is $Y = 200X - 500$. Using the linearity of expectation, compute $E(Y)$. Is the kiosk viable on average?

11. Using $\text{Var}(aX + b) = a^2\text{Var}(X)$, compute $\text{Var}(Y)$ and $\sigma(Y)$ (to 2 decimal places).
12. Compute $P(Y > 0)$, the probability that a randomly chosen hour is profitable, using the c.d.f.
13. Write a short advisory note (three to five sentences) to Madame Etongwe: is the kiosk viable, how large is the risk, and what concrete change (e.g. negotiating a higher commission or reducing hourly costs) would make the expected profit positive?

**Part E — Extension: the moment generating function**

14. Write down the m.g.f $M_X(t) = E\left(e^{tX}\right)$ of $X$ as a finite sum.
15. Compute $M_X'(t)$ and verify that $M_X'(0) = E(X)$.
16. Compute $M_X''(t)$ and verify that $M_X''(0) = E(X^2)$, then recover $\text{Var}(X)$.

## 7.4 Model Answer

**Part A — The probability distribution.**

**1.** The p.m.f of $X$ is:

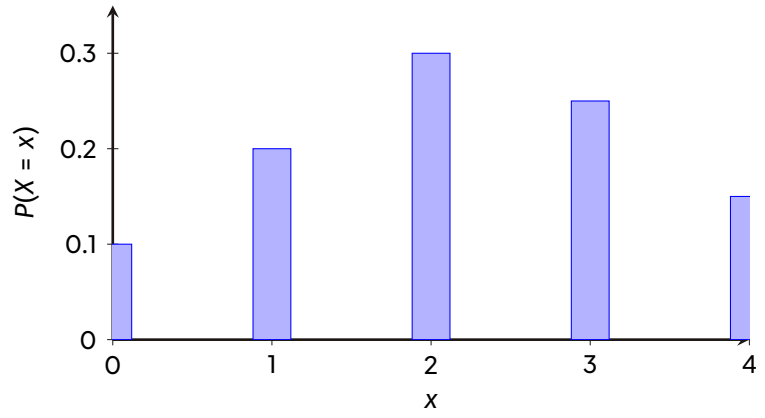
| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(X = x)$ | 0.10 | 0.20 | 0.30 | 0.25 | 0.15 |

**2.** A table defines a valid probability distribution if (i) every probability satisfies $0 \le P(X = x) \le 1$ and (ii) $\sum_x P(X = x) = 1$. Here all five values lie in $[0, 1]$, and

$$0.10 + 0.20 + 0.30 + 0.25 + 0.15 = 1.00.$$

Both conditions hold, so the table is a valid p.m.f.

**3.** Bar chart of the p.m.f:




Probability mass function of the hourly number of withdrawals $X$.

**Part B — Expectation and variance.**

**4.** By definition, $E(X) = \sum_x x\,P(X = x)$:

$$E(X) = 0(0.10) + 1(0.20) + 2(0.30) + 3(0.25) + 4(0.15)$$
$$= 0 + 0.20 + 0.60 + 0.75 + 0.60 = 2.15.$$

Interpretation: over a long period, the agent serves on average 2.15 withdrawal customers per hour.

**5.** First compute $E(X^2) = \sum_x x^2 P(X = x)$:

$$E(X^2) = 0(0.10) + 1(0.20) + 4(0.30) + 9(0.25) + 16(0.15)$$
$$= 0 + 0.20 + 1.20 + 2.25 + 2.40 = 6.05.$$

Then

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = 6.05 - (2.15)^2 = 6.05 - 4.6225 = 1.4275.$$

**6.** $\sigma(X) = \sqrt{1.4275} \approx 1.19$ customers. The hourly number of withdrawals typically fluctuates by about 1.2 customers around the mean of 2.15: the flow is fairly variable relative to its average.

**Part C — The c.d.f, mode and median.**

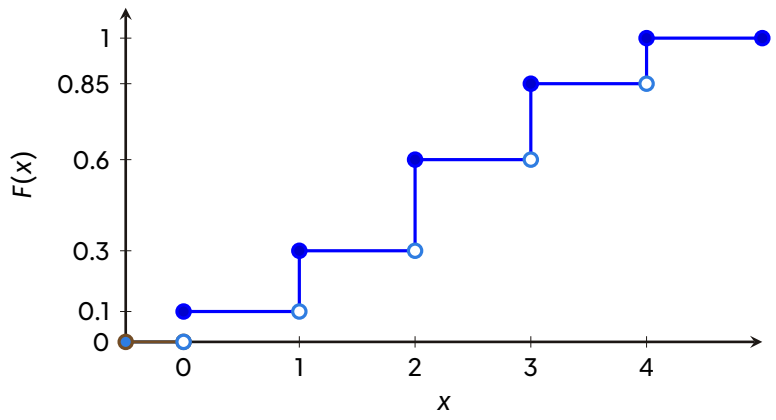
**7.** Accumulating the probabilities:

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $F(x) = P(X \le x)$ | 0.10 | 0.30 | 0.60 | 0.85 | 1.00 |

As a piecewise function:

$$F(x) = \begin{cases} 0 & \text{if } x < 0, \\ 0.10 & \text{if } 0 \le x < 1, \\ 0.30 & \text{if } 1 \le x < 2, \\ 0.60 & \text{if } 2 \le x < 3, \\ 0.85 & \text{if } 3 \le x < 4, \\ 1 & \text{if } x \ge 4. \end{cases}$$

**8.** Graph of $F$:




Cumulative distribution function of $X$ (closed dots included, open dots excluded).

**9.** Mode: the mode is the value with the largest probability. Since $P(X = 2) = 0.30$ is the greatest entry, the mode is $\boxed{2}$ withdrawals per hour. Median: the median is the smallest value $m$ such that $F(m) \geq 0.5$. Here $F(1) = 0.30 < 0.5$ and $F(2) = 0.60 \geq 0.5$, so the median is $\boxed{2}$. In half of the hours, at most 2 withdrawal customers arrive.

**Part D — Profit analysis.**

**10.** With $Y = 200X - 500$:

$$E(Y) = 200\,E(X) - 500 = 200(2.15) - 500 = 430 - 500 = -70 \text{ FCFA}.$$

On average, the kiosk loses 70 FCFA per hour: as things stand, it is **not viable**.

**11.** $\operatorname{Var}(Y) = 200^2 \operatorname{Var}(X) = 40\,000 \times 1.4275 = 57\,100 \text{ FCFA}^2$, so

$$\sigma(Y) = \sqrt{57\,100} \approx 238.96 \text{ FCFA}.$$

The hourly profit fluctuates by roughly 239 FCFA around its (negative) mean: the risk is large compared with the average loss.

**12.** $Y > 0 \iff 200X > 500 \iff X \geq 3$. Hence

$$P(Y > 0) = P(X \geq 3) = 1 - F(2) = 1 - 0.60 = 0.40.$$

Only 40% of hours are profitable.

**13.** Advisory note. Madame, your kiosk serves on average 2.15 withdrawal customers per hour, giving an expected hourly income of 430 FCFA against fixed costs of 500 FCFA: you lose on average 70 FCFA every hour, and only 4 hours out of 10 are profitable. The standard deviation of your profit, about 239 FCFA, shows that good hours cannot be relied upon to compensate. To break even you need $E(X) \geq 2.5$ customers per hour; we advise you either to negotiate a commission of at least $\frac{500}{2.15} \approx 233$ FCFA per withdrawal, or to cut hourly costs by at least 70 FCFA, or to attract about 0.35 more customers per hour through advertising.

**Part E — Extension: the m.g.f.**

**14.** $M_X(t) = E(e^{tX}) = \sum_x e^{tx} P(X = x)$:

$$M_X(t) = 0.10 + 0.20e^{t} + 0.30e^{2t} + 0.25e^{3t} + 0.15e^{4t}.$$

**15.** Differentiating term by term:

$$M_X'(t) = 0.20e^{t} + 0.60e^{2t} + 0.75e^{3t} + 0.60e^{4t},$$

$$M_X'(0) = 0.20 + 0.60 + 0.75 + 0.60 = 2.15 = E(X). \checkmark$$

**16.** Differentiating again:

$$M_X''(t) = 0.20e^{t} + 1.20e^{2t} + 2.25e^{3t} + 2.40e^{4t},$$

$$M_X''(0) = 0.20 + 1.20 + 2.25 + 2.40 = 6.05 = E(X^2). \checkmark$$

Therefore

$$\operatorname{Var}(X) = M_X''(0) - \left[M_X'(0)\right]^2 = 6.05 - (2.15)^2 = 1.4275,$$

exactly as found in Part B.

**WARNING** **Common mistakes to avoid in your presentation**

- Forgetting to square $x$ when computing $E(X^2)$: $x^2P(X = x)$, not $xP(X = x)$.
- Writing $\text{Var}(aX + b) = a^2\text{Var}(X) + b$: the constant $b$ shifts the profit but does not change its variance.
- Reading the median from the p.m.f instead of the c.d.f: the median is defined through $F(m) \geq 0.5$.
- Concluding "viable" because $P(Y > 0) = 0.40$ is "almost half": viability is judged by $E(Y)$, and here $E(Y) = -70$ FCFA.
- In the m.g.f, differentiating $e^{tx}$ with respect to $x$ instead of $t$.

**DID YOU KNOW?** **Why this matters beyond the classroom**

Mobile money agents across Cameroon — in Buea, Douala, Bamenda and Yaoundé — face exactly this trade-off between commission income and fixed operating costs. Modelling the hourly customer flow as a discrete random variable, and summarising it by its expectation and variance, is precisely how operators and agents decide where to open kiosks and how much float to hold. The same mathematics drives inventory planning, insurance pricing and call-centre staffing worldwide.

# 8 METHODS AND WORKED EXAMPLES

# 9 DISCRETE RANDOM VARIABLES

## 9.1 Discrete random variables and probability distributions

**DEFINITION** **Discrete random variable**

A **discrete random variable** $X$ is a function that assigns a real number to each outcome of a random experiment, where the set of possible values is either finite or countably infinite. We denote the values by $x_1, x_2, \ldots$ and the corresponding probabilities by $P(X = x_i)$.

**DEFINITION** **Probability mass function (p.m.f.)**

The **probability mass function** (p.m.f.) of a discrete random variable $X$ is the function $p(x) = P(X = x)$ satisfying:

1. $p(x) \geq 0$ for all $x$,
2. $\sum_x p(x) = 1$, where the sum is over all possible values of $X$.

The collection of pairs $(x, p(x))$ is called the **probability distribution** of $X$.

**WARNING** **p.m.f. vs p.d.f.**

The term probability density function (p.d.f.) is reserved for continuous random variables. For discrete variables we always say probability mass function (p.m.f.). The syllabus mentions both names; in examinations, use p.m.f. for discrete distributions.

**METHOD** **Determining the p.m.f. of a discrete random variable**

1. Identify the **sample space** of the experiment and list all possible values $x_1, x_2, \ldots$ that $X$ can take (count carefully — use a table of outcomes or a tree diagram if needed).
2. For each value $x$, compute $P(X = x)$ by counting equally likely favourable outcomes or by multiplying probabilities along a tree.

3. Present the results in a **probability distribution table**:

| $x$ | $x_1$ | $x_2$ | … | $x_n$ |
|---|---|---|---|---|
| $P(X = x)$ | $p_1$ | $p_2$ | … | $p_n$ |

4. Check: every $p_i \geq 0$ and $\sum p_i = 1$. If an unknown constant $k$ appears, use $\sum p_i = 1$ to find it.

Reflex: if probabilities are given as expressions in $k$, write the equation $\sum P(X = x) = 1$ immediately — it is almost always the key to the first marks.

**WORKED EXAMPLE 9** **Finding an unknown constant and probabilities**

A discrete random variable $X$ has probability distribution

$$P(X = x) = \begin{cases} kx, & x = 1, 2, 3, 4, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the value of $k$. (b) Find $P(X \geq 3)$ and $P(1 < X \leq 3)$.

**SOLUTION**

(a) The values taken by $X$ are $1, 2, 3, 4$ with probabilities $k, 2k, 3k, 4k$. Since the sum of all probabilities equals 1:

$$k + 2k + 3k + 4k = 1 \implies 10k = 1 \implies k = \frac{1}{10}.$$

The distribution is therefore

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $P(X = x)$ | 0.1 | 0.2 | 0.3 | 0.4 |

(b) $P(X \geq 3) = P(X = 3) + P(X = 4) = 0.3 + 0.4 = 0.7$.
$P(1 < X \leq 3) = P(X = 2) + P(X = 3) = 0.2 + 0.3 = 0.5$.
Check: probabilities are all non-negative and sum to $0.1 + 0.2 + 0.3 + 0.4 = 1$. ✓

## 9.2 The probability mass function and expectation

**DEFINITION** **Expectation (mean) of a discrete random variable**

The **expectation** (or **mean**) of a discrete random variable $X$ is

$$E(X) = \mu = \sum_{x} x\,P(X = x),$$

provided the series converges absolutely. It represents the long-run average value of $X$ over many independent repetitions of the experiment.

KEY POINTS **Law of the unconscious statistician**

For any function $g$,

$$E(g(X)) = \sum_{x} g(x)\,P(X = x).$$

Do not compute the distribution of $g(X)$ first — work directly with the table of $X$.

PROPERTY **Linearity of expectation**

For any constants $a, b \in \mathbb{R}$,

$$E(aX + b) = aE(X) + b.$$

If $X$ and $Y$ are any two random variables (independent or not),

$$E(X + Y) = E(X) + E(Y).$$

METHOD **Expectation of $X$ and of a function of $X$**

1. Write the distribution table of $X$.
2. Compute the **expectation** (mean):

$$E(X) = \mu = \sum_{x} x\,P(X = x).$$

3. For a function $g(X)$, use the **law of the unconscious statistician**:

$$E(g(X)) = \sum_{x} g(x)\,P(X = x).$$

4. Remember linearity: $E(aX + b) = aE(X) + b$.

Reflex: $E(X)$ is a weighted average; it need not be one of the values of $X$.

**WORKED EXAMPLE 10** **Expectation in a game of chance**

A stall at the Yaoundé annual trade fair offers the following game. A player pays 500 FCFA and spins a wheel. The wheel lands on 0, 1, 2 or 3 with probabilities 0.5, 0.3, 0.15 and 0.05 respectively, and the player receives $1000x$ FCFA, where $x$ is the number shown. Let $G$ be the player's net gain in FCFA. Find $E(G)$ and state whether the game is fair.

**SOLUTION**

The gross gain is $1000X$ and the player paid 500, so $G = 1000X - 500$.
Step 1: compute $E(X)$.

$$E(X) = 0(0.5) + 1(0.3) + 2(0.15) + 3(0.05) = 0 + 0.3 + 0.3 + 0.15 = 0.75.$$

Step 2: use linearity of expectation.

$$E(G) = E(1000X - 500) = 1000\,E(X) - 500 = 1000(0.75) - 500 = 250.$$

Conclusion: $E(G) = 250$ FCFA $> 0$, so the game is not fair; it is favourable to the player, who gains on average 250 FCFA per game in the long run. (A fair game would require $E(G) = 0$, i.e. a stake of 750 FCFA.)

## 9.3 Variance and standard deviation of a discrete random variable

**DEFINITION** **Variance and standard deviation**

The **variance** of $X$ measures the spread of the distribution about the mean:

$$\operatorname{Var}(X) = E\left[(X - \mu)^2\right] = \sum_{x} (x - \mu)^2 P(X = x).$$

The **standard deviation** is $\sigma = \sqrt{\operatorname{Var}(X)}$ (same units as $X$).

**PROPERTY** **Computational formula and properties**

$$\operatorname{Var}(X) = E(X^2) - [E(X)]^2.$$

For constants $a$, $b$:

$$\operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X).$$

If $X$ and $Y$ are independent:

$$\operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y).$$

**WARNING**

Never write $\text{Var}(aX+b) = a\,\text{Var}(X)+b$ — this is a classic GCE error. The constant $b$ disappears and $a$ is squared. Also $\text{Var}(X) \geq 0$ always; a negative answer signals an arithmetic slip.

**METHOD** **Variance** $\text{Var}(X)$

1. Compute $\mu = E(X)$.
2. Compute $E(X^2) = \sum_x x^2 P(X = x)$ — build a row of $x^2$ values under the distribution table.
3. Apply the **computational formula**:

$$\text{Var}(X) = E(X^2) - [E(X)]^2.$$

4. The **standard deviation** is $\sigma = \sqrt{\text{Var}(X)}$.
5. Properties to remember: $\text{Var}(aX + b) = a^2\,\text{Var}(X)$ (adding a constant changes nothing); if $X$ and $Y$ are independent, $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.

**WORKED EXAMPLE 11** **Mean, variance and a linear transformation**

A discrete random variable $X$ has distribution

| $x$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|
| $P(X = x)$ | $0.2$ | $0.3$ | $0.4$ | $0.1$ |

(a) Find $E(X)$ and $\text{Var}(X)$. (b) If $Y = 3X - 2$, find $E(Y)$, $\text{Var}(Y)$ and the standard deviation of $Y$.

**SOLUTION**

(a) Mean:

$$E(X) = (-1)(0.2) + 0(0.3) + 1(0.4) + 2(0.1) = -0.2 + 0 + 0.4 + 0.2 = 0.4.$$

Second moment: the squares of the values are $1, 0, 1, 4$, so

$$E(X^2) = 1(0.2) + 0(0.3) + 1(0.4) + 4(0.1) = 0.2 + 0 + 0.4 + 0.4 = 1.0.$$

Variance:

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 1.0 - (0.4)^2 = 1.0 - 0.16 = 0.84.$$

(b) Using $E(aX + b) = aE(X) + b$ and $\text{Var}(aX + b) = a^2\text{Var}(X)$ with $a = 3$, $b = -2$:

$$E(Y) = 3(0.4) - 2 = -0.8, \qquad \text{Var}(Y) = 3^2 \times 0.84 = 9 \times 0.84 = 7.56.$$

$$\sigma_Y = \sqrt{7.56} \approx 2.75 \quad (3\text{ s.f.})$$

Remark: subtracting 2 shifts every value but does not spread them out — only the factor 3 affects the variance, through $3^2$.

## 9.4 The cumulative distribution function, mode and median

**DEFINITION** **Cumulative distribution function (c.d.f.)**

The **cumulative distribution function** (c.d.f.) of a discrete random variable $X$ is

$$F(x) = P(X \le x) = \sum_{x_i \le x} P(X = x_i), \quad x \in \mathbb{R}.$$

**PROPERTY** **Properties of the c.d.f.**

For a discrete random variable:

1. $F$ is non-decreasing: if $a < b$ then $F(a) \le F(b)$.
2. $F$ is right-continuous: $F(x) = \lim_{h \to 0^+} F(x + h)$.
3. $\lim_{x \to -\infty} F(x) = 0$, $\lim_{x \to +\infty} F(x) = 1$.
4. $P(X = x_i) = F(x_i) - F(x_i^-)$ (the jump at $x_i$).
5. $P(a < X \le b) = F(b) - F(a)$.

**METHOD** **The cumulative distribution function $F(x)$**

1. Order the values of $X$: $x_1 < x_2 < \cdots < x_n$.
2. The **c.d.f.** is $F(x) = P(X \le x) = \sum_{x_i \le x} p_i$: accumulate probabilities from left to right.
3. Write $F$ as a **step function**:

$$F(x) = \begin{cases} 0, & x < x_1, \\ p_1, & x_1 \le x < x_2, \\ p_1 + p_2, & x_2 \le x < x_3, \\ \vdots & \\ 1, & x \ge x_n. \end{cases}$$

4. To recover the p.m.f. from $F$: $P(X = x_i) = F(x_i) - F(x_{i-1})$ (the jump at $x_i$).
5. Useful: $P(X > x) = 1 - F(x)$ and $P(a < X \le b) = F(b) - F(a)$.

Reflex: $F$ is non-decreasing, right-continuous, with $F \to 0$ on the far left and $F \to 1$ on the far right.

**WORKED EXAMPLE 12** **From p.m.f. to c.d.f. and back**

The number $X$ of goals scored per match by a school team in Bamenda has distribution

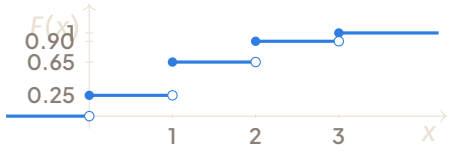
| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $P(X = x)$ | 0.25 | 0.40 | 0.25 | 0.10 |

(a) Construct the c.d.f. $F(x)$ and sketch its graph. (b) Use $F$ to find $P(X \le 1)$, $P(X > 1)$ and $P(0 < X \le 2)$. (c) Verify that $F(2) - F(1)$ gives $P(X = 2)$.

**SOLUTION**

(a) Accumulate the probabilities: 0.25; 0.25+0.40 = 0.65; 0.65+0.25 = 0.90; 0.90+0.10 = 1.00. Hence

$$F(x) = \begin{cases} 0, & x < 0, \\ 0.25, & 0 \le x < 1, \\ 0.65, & 1 \le x < 2, \\ 0.90, & 2 \le x < 3, \\ 1, & x \ge 3. \end{cases}$$




Step graph of the c.d.f.: closed dots on the left of each step (right-continuity).

(b) $P(X \le 1) = F(1) = 0.65$.
$P(X > 1) = 1 - F(1) = 1 - 0.65 = 0.35$.
$P(0 < X \le 2) = F(2) - F(0) = 0.90 - 0.25 = 0.65$ (indeed $P(X = 1) + P(X = 2) = 0.40 + 0.25$).
(c) $F(2)-F(1) = 0.90-0.65 = 0.25 = P(X = 2)$. ✓ The jump of $F$ at $x = 2$ is exactly the probability mass there.

**DEFINITION** **Mode and median**

The **mode** of a discrete random variable is any value $x$ for which $P(X = x)$ is maximal. There may be more than one mode (bimodal, multimodal).
A **median** $m$ is any value satisfying

$$P(X \le m) \ge \tfrac{1}{2} \quad \text{and} \quad P(X \ge m) \ge \tfrac{1}{2}.$$

Equivalently, the median is the smallest $x$ such that $F(x) \ge 0.5$.

**METHOD** **Mode and median of a discrete random variable**

1. Mode: scan the p.m.f.; the **mode** is any value $x$ with the largest probability. There may be two modes (bimodal).
2. Median: compute the c.d.f. $F$. A **median** $m$ is any value such that

   $$P(X \le m) \ge \tfrac{1}{2} \quad \text{and} \quad P(X \ge m) \ge \tfrac{1}{2}.$$

   Equivalently, find the smallest $x$ with $F(x) \ge 0.5$.
3. If $F$ jumps from below 0.5 to above 0.5 at $x_k$, the median is $x_k$. If $F(x_k) = 0.5$ exactly, every value in $[x_k, x_{k+1}]$ is a median (by GCE convention we usually quote the interval or the smallest value, as required).

Reflex: mode = highest bar of the p.m.f.; median = where the c.d.f. crosses 0.5.

**WORKED EXAMPLE 13** **Mode and median**

The number $X$ of candidates absent per day in an Upper Sixth class has distribution

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(X = x)$ | 0.10 | 0.35 | 0.30 | 0.15 | 0.10 |

Find (a) the mode, (b) the median, (c) the mean, and comment briefly.

**SOLUTION**

(a) Mode: the largest probability is 0.35 at $x = 1$. Hence mode = 1.
(b) Median: build the c.d.f.:

$$F(0) = 0.10, \quad F(1) = 0.45, \quad F(2) = 0.75, \quad F(3) = 0.90, \quad F(4) = 1.$$

The smallest $x$ with $F(x) \ge 0.5$ is $x = 2$ (since $F(1) = 0.45 < 0.5$ and $F(2) = 0.75 \ge 0.5$). Check

both conditions:

$$P(X \le 2) = 0.75 \ge \tfrac{1}{2}, \qquad P(X \ge 2) = 0.30 + 0.15 + 0.10 = 0.55 \ge \tfrac{1}{2}.$$

Hence the median is $m = 2$.
(c) Mean:

$$E(X) = 0(0.10) + 1(0.35) + 2(0.30) + 3(0.15) + 4(0.10) = 0.35 + 0.60 + 0.45 + 0.40 = 1.80.$$

Comment: mode (1) < median (2); the mean (1.8) lies between them, pulled slightly above the mode by the right tail — typical of a distribution skewed to the right.

## 9.5 Moment generating functions

**DEFINITION** **Moment generating function (m.g.f.)**

The **moment generating function** of a discrete random variable $X$ is

$$M_X(t) = E\left(e^{tX}\right) = \sum_x e^{tx}\, P(X = x),$$

defined for all $t$ in an interval containing 0 where the sum converges. Always $M_X(0) = 1$.

**PROPERTY** **Properties of the m.g.f.**

1. Moments: $E(X) = M_X'(0)$, $E(X^2) = M_X''(0)$, and in general $E(X^n) = M_X^{(n)}(0)$.
2. Variance: $\mathrm{Var}(X) = M_X''(0) - [M_X'(0)]^2$.
3. Uniqueness: the m.g.f. uniquely determines the distribution (if it exists in a neighbourhood of 0).
4. Sum of independent variables: if $X$ and $Y$ are independent, $M_{X+Y}(t) = M_X(t)\, M_Y(t)$.
5. Linear transformation: $M_{aX+b}(t) = e^{bt} M_X(at)$.

**METHOD** **Moment generating function $M_X(t)$**

1. The **moment generating function** is

$$M_X(t) = E\left(e^{tX}\right) = \sum_x e^{tx}\, P(X = x),$$

defined for all $t$ where this sum converges (always at $t = 0$, with $M_X(0) = 1$).

2. Extracting moments: differentiate and evaluate at $t = 0$:

$$E(X) = M_X'(0), \qquad E(X^2) = M_X''(0), \qquad \mathrm{Var}(X) = M_X''(0) - \left[M_X'(0)\right]^2.$$

3. Recovering the distribution: expand $M_X(t)$ as a sum of exponentials; the coefficient of $e^{tx}$ is $P(X = x)$.

4. Sums of independent variables: if $X$ and $Y$ are independent,

$$M_{X+Y}(t) = M_X(t)\, M_Y(t).$$

Reflex: the m.g.f. packages all moments in one function — differentiate, then set $t = 0$.

**WORKED EXAMPLE 14** **M.g.f. of a small distribution**

A random variable $X$ takes the value 0 with probability $\frac{1}{3}$ and the value 2 with probability $\frac{2}{3}$. (a) Find $M_X(t)$. (b) Use $M_X$ to find $E(X)$ and $\mathrm{Var}(X)$. (c) Verify directly from the distribution.

**SOLUTION**

(a) By definition,

$$M_X(t) = E(e^{tX}) = \tfrac{1}{3}e^0 + \tfrac{2}{3}e^{2t} = \tfrac{1}{3} + \tfrac{2}{3}e^{2t}.$$

(b) Differentiate with respect to $t$:

$$M_X'(t) = \tfrac{2}{3}\cdot 2e^{2t} = \tfrac{4}{3}e^{2t} \implies E(X) = M_X'(0) = \tfrac{4}{3}.$$

$$M_X''(t) = \tfrac{4}{3}\cdot 2e^{2t} = \tfrac{8}{3}e^{2t} \implies E(X^2) = M_X''(0) = \tfrac{8}{3}.$$

Hence

$$\mathrm{Var}(X) = M_X''(0) - [M_X'(0)]^2 = \tfrac{8}{3} - \left(\tfrac{4}{3}\right)^2 = \tfrac{8}{3} - \tfrac{16}{9} = \tfrac{24-16}{9} = \tfrac{8}{9}.$$

(c) Direct check:

$$E(X) = 0\cdot\tfrac{1}{3} + 2\cdot\tfrac{2}{3} = \tfrac{4}{3}, \qquad E(X^2) = 0\cdot\tfrac{1}{3} + 4\cdot\tfrac{2}{3} = \tfrac{8}{3},$$

$$\mathrm{Var}(X) = \tfrac{8}{3} - \tfrac{16}{9} = \tfrac{8}{9}. \quad \checkmark$$

The two methods agree, as they must.

## 9.6 Synoptic GCE-style problem

**METHOD** **Tackling a full examination question**

1. Read twice and define $X$ precisely; list its values.
2. Use $\sum p_i = 1$ for any unknown constant.
3. Answer each part with the right tool: table → p.m.f.; accumulated sums → c.d.f.; $\sum x p_i$, $\sum x^2 p_i$ → mean and variance; highest bar → mode; crossing of 0.5 → median; $E(e^{tX})$ → m.g.f.
4. Keep exact fractions as long as possible; round only at the end, to the accuracy demanded.
5. Conclude each part with a sentence in the context of the question.

**WORKED EXAMPLE 15** **Comprehensive problem — GCE style**

A bag contains 3 red and 2 blue marbles. Two marbles are drawn at random without replacement. Let $X$ be the number of red marbles drawn. (a) Show that the probability distribution of $X$ is given by

$$P(X=0)=\frac{1}{10},\qquad P(X=1)=\frac{6}{10},\qquad P(X=2)=\frac{3}{10}.$$

(b) Write down the c.d.f. $F(x)$ of $X$. (c) Find the mode and the median of $X$. (d) Calculate $E(X)$ and $\operatorname{Var}(X)$. (e) Find the moment generating function $M_X(t)$ and verify that $M_X'(0)=E(X)$. (f) Two independent repetitions of the experiment are performed, giving $X_1$ and $X_2$. Write down $M_{X_1+X_2}(t)$.

**SOLUTION**

(a) Total marbles: 5. The number of ways of choosing 2 marbles from 5 is $\binom{5}{2}=10$.

$$P(X=0)=\frac{\binom{3}{0}\binom{2}{2}}{\binom{5}{2}}=\frac{1\times 1}{10}=\frac{1}{10},$$

$$P(X=1)=\frac{\binom{3}{1}\binom{2}{1}}{\binom{5}{2}}=\frac{3\times 2}{10}=\frac{6}{10},\qquad P(X=2)=\frac{\binom{3}{2}\binom{2}{0}}{\binom{5}{2}}=\frac{3\times 1}{10}=\frac{3}{10}.$$

Check: $\frac{1}{10}+\frac{6}{10}+\frac{3}{10}=1$. ✓

(b) Accumulating:

$$F(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{10}, & 0 \le x < 1, \\ \frac{7}{10}, & 1 \le x < 2, \\ 1, & x \ge 2. \end{cases}$$

(c) Mode: the largest probability is $\frac{6}{10}$ at $x = 1$, so the mode is 1.
Median: $F(0) = \frac{1}{10} < 0.5$ and $F(1) = \frac{7}{10} \ge 0.5$; also $P(X \ge 1) = \frac{9}{10} \ge \frac{1}{2}$. Hence the median is 1.
(d)

$$E(X) = 0 \cdot \tfrac{1}{10} + 1 \cdot \tfrac{6}{10} + 2 \cdot \tfrac{3}{10} = \tfrac{12}{10} = \tfrac{6}{5} = 1.2.$$

$$E(X^2) = 0 \cdot \tfrac{1}{10} + 1 \cdot \tfrac{6}{10} + 4 \cdot \tfrac{3}{10} = \tfrac{18}{10} = \tfrac{9}{5}.$$

$$\mathrm{Var}(X) = \tfrac{9}{5} - \left(\tfrac{6}{5}\right)^2 = \tfrac{9}{5} - \tfrac{36}{25} = \tfrac{45-36}{25} = \tfrac{9}{25} = 0.36.$$

(e)

$$M_X(t) = E(e^{tX}) = \tfrac{1}{10}e^0 + \tfrac{6}{10}e^t + \tfrac{3}{10}e^{2t} = \tfrac{1}{10}\left(1 + 6e^t + 3e^{2t}\right).$$

Differentiate:

$$M_X'(t) = \tfrac{1}{10}\left(6e^t + 6e^{2t}\right) \implies M_X'(0) = \tfrac{1}{10}(6 + 6) = \tfrac{12}{10} = \tfrac{6}{5} = E(X). \quad \checkmark$$

(f) Since $X_1$ and $X_2$ are independent, the m.g.f. of the sum is the product:

$$M_{X_1+X_2}(t) = M_{X_1}(t)\,M_{X_2}(t) = \left[\tfrac{1}{10}\left(1 + 6e^t + 3e^{2t}\right)\right]^2 = \tfrac{1}{100}\left(1 + 6e^t + 3e^{2t}\right)^2.$$

Contextual remark: on average 1.2 red marbles are drawn, the most likely outcome is exactly one red, and the spread about the mean is $\sigma = \sqrt{0.36} = 0.6$ marble.

# 10 EXERCISES

## 10.1 I check my knowledge

**EXERCISE 5** **Multiple choice**

A discrete random variable $X$ takes the values $0, 1, 2$ with probabilities $0.3$, $0.5$ and $0.2$ respectively. The value of $E(X)$ is:

1. A) $0.8$
2. B) $0.9$
3. C) $1.0$
4. D) $1.1$

**EXERCISE 6** **True or false — justify**

For each statement, say whether it is true or false and justify your answer.

1. If $p(x)$ is the p.m.f of a discrete random variable $X$, then $\sum_x p(x) = 1$.
2. For any discrete random variable $X$, $\mathrm{Var}(X) = E(X^2) - E(X)$.
3. The cumulative distribution function $F(x)$ of a discrete random variable is always an increasing function of $x$.
4. If $M_X(t)$ is the moment generating function of $X$, then $E(X) = M_X'(0)$.

**EXERCISE 7** **Definitions**

Give a precise definition of each of the following, illustrating each one with a small example:

1. the **probability mass function** of a discrete random variable $X$;
2. the **cumulative distribution function** of $X$;
3. the **mode** and the **median** of a discrete random variable.

**EXERCISE 8** **Direct calculations**

Given that $E(X) = 4$ and $\text{Var}(X) = 9$, find:

1. $E(3X - 2)$;
2. $\text{Var}(3X - 2)$;
3. $E(X^2)$;
4. $\text{Var}(5 - 2X)$.

## 10.2 I practise

**EXERCISE 9** **Finding the constant of a p.m.f**

A discrete random variable $X$ has p.m.f

$$p(x) = kx \quad \text{for } x = 1, 2, 3, 4, \qquad p(x) = 0 \text{ otherwise.}$$

1. Find the value of the constant $k$.
2. Compute $E(X)$ and $\text{Var}(X)$.
3. Determine the mode and the median of $X$.
4. Sketch the cumulative distribution function $F(x)$.

**EXERCISE 10** **From the c.d.f to the p.m.f**

The cumulative distribution function of a discrete random variable $X$ is given by

$$F(x) = \begin{cases} 0 & x < 1 \\ 0.2 & 1 \le x < 2 \\ 0.5 & 2 \le x < 3 \\ 0.8 & 3 \le x < 4 \\ 1 & x \ge 4 \end{cases}$$

1. Determine the p.m.f of $X$ and present it in a table.
2. Compute $E(X)$ and $\text{Var}(X)$.
3. Find the median of $X$.

**EXERCISE 11** **A fair game?**

A game at a school fair in Buea costs 100 FCFA to play. A player wins 500 FCFA with probability $\frac{1}{10}$, wins 200 FCFA with probability $\frac{1}{5}$, and wins nothing otherwise.

1. Let $G$ be the player's net gain (winnings minus the stake). Write down the probability distribution of $G$.
2. Compute the expected gain $E(G)$ and the standard deviation of $G$.
3. Is the game fair? If not, find the stake that would make it fair.

**EXERCISE 12** **Linear combinations of independent variables**

Two independent random variables $X$ and $Y$ satisfy

$$E(X) = 2, \quad \mathrm{Var}(X) = 1, \quad E(Y) = 5, \quad \mathrm{Var}(Y) = 4.$$

1. Find $E(2X + 3Y)$.
2. Find $\mathrm{Var}(2X + 3Y)$, stating clearly at which step the independence of $X$ and $Y$ is used.
3. Find also $E(X - Y)$ and $\mathrm{Var}(X - Y)$.

## 10.3 I prepare for the GCE

**EXERCISE 13** **Moment generating function of a Bernoulli variable** **(10 marks)**

A random variable $X$ takes the value 1 with probability $p$ and the value 0 with probability $1 - p$, where $0 < p < 1$.

1. Show that the moment generating function of $X$ is $M(t) = (1 - p) + pe^{t}$. (3 marks)
2. By differentiating $M(t)$, derive $E(X)$ and $E(X^2)$. (4 marks)
3. Hence show that $\mathrm{Var}(X) = p(1 - p)$. (3 marks)

**EXERCISE 14** **Recovering a distribution from its moments** **(12 marks)**

A random variable $X$ takes the values 1, 2, 3 with probabilities $a$, $b$, $c$ respectively. It is given that

$$E(X) = 2 \qquad \text{and} \qquad E(X^2) = \frac{14}{3}.$$

1. Set up three equations satisfied by $a$, $b$ and $c$. (3 marks)

2. Solve these equations to find $a$, $b$ and $c$. (4 marks)
3. Determine the median and the mode of $X$. (2 marks)
4. Construct the cumulative distribution function $F(x)$ of $X$ and sketch its graph. (3 marks)

**EXERCISE 15** **Defective items in a sample — GCE style** **(16 marks)**

A trader at Mokolo market receives items in batches. The number $X$ of defective items in a random sample of 3 items has the following probability distribution:

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $P(X = x)$ | 0.512 | $k$ | 0.096 | 0.008 |

1. Find the value of the constant $k$. (2 marks)
2. Compute $E(X)$ and $\mathrm{Var}(X)$. (5 marks)
3. Construct the cumulative distribution function $F(x)$ and sketch its graph. (4 marks)
4. State the mode and the median of $X$. (2 marks)
5. Write down the moment generating function $M_X(t)$ of $X$ and use it to verify the value of $E(X)$ found in part 2. (3 marks)

# 11 SOLUTIONS TO THE EXERCISES

**SOLUTION** **Exercise 1 — Multiple choice**

The expectation of a discrete random variable is $E(X) = \sum x\,P(X = x)$.

$$E(X) = 0 \times 0.3 + 1 \times 0.5 + 2 \times 0.2 = 0 + 0.5 + 0.4 = 0.9.$$

The correct answer is B) 0.9.

**SOLUTION** **Exercise 2 — True or false — justify**

1. True. By definition, the probability mass function $p(x) = P(X = x)$ satisfies $\sum_x p(x) = 1$ because the probabilities of all possible outcomes must sum to 1.
2. False. The correct formula is $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$. The given expression misses the square on $E(X)$.
3. False. The cumulative distribution function $F(x) = P(X \le x)$ is non-decreasing (it never decreases), but it is not strictly increasing; it is constant between possible values of $X$ and jumps at each value.
4. True. The moment generating function is $M_X(t) = E(e^{tX})$. Differentiating under the expectation gives $M_X'(t) = E(Xe^{tX})$, so $M_X'(0) = E(X)$.

**SOLUTION** **Exercise 3 — Definitions**

1. Probability mass function (p.m.f) of a discrete random variable $X$ is the function $p(x) = P(X = x)$ for each possible value $x$. It satisfies $p(x) \ge 0$ for all $x$ and $\sum_x p(x) = 1$.
   Example: Let $X$ be the outcome of a fair die. Then $p(x) = \frac{1}{6}$ for $x = 1, 2, 3, 4, 5, 6$ and 0 otherwise.
2. Cumulative distribution function (c.d.f) of $X$ is $F(x) = P(X \le x)$ for all real $x$. It is non-decreasing, right-continuous, with $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to+\infty} F(x) = 1$.
   Example: For the fair die, $F(x) = 0$ for $x < 1$, $\frac{1}{6}$ for $1 \le x < 2$, $\frac{2}{6}$ for $2 \le x < 3$, …, 1 for $x \ge 6$.
3. Mode of $X$ is the value $x$ that maximises $p(x)$ (the most probable value). Median is any value $m$ such that $P(X \le m) \ge \frac{1}{2}$ and $P(X \ge m) \ge \frac{1}{2}$; for a discrete variable we usually take the smallest such $m$.
   Example: If $P(X = 1) = 0.2$, $P(X = 2) = 0.5$, $P(X = 3) = 0.3$, the mode is 2 and the median is 2 (since $P(X \le 2) = 0.7 \ge 0.5$ and $P(X \ge 2) = 0.8 \ge 0.5$).

**SOLUTION** **Exercise 4 — Direct calculations**

Given $E(X) = 4$ and $\mathrm{Var}(X) = 9$.

1. $E(3X - 2) = 3E(X) - 2 = 3 \times 4 - 2 = 10$.
2. $\mathrm{Var}(3X - 2) = 3^2\,\mathrm{Var}(X) = 9 \times 9 = 81$.
3. $E(X^2) = \mathrm{Var}(X) + [E(X)]^2 = 9 + 16 = 25$.
4. $\mathrm{Var}(5 - 2X) = (-2)^2\,\mathrm{Var}(X) = 4 \times 9 = 36$.

**SOLUTION** **Exercise 5 — Finding the constant of a p.m.f**

1. The p.m.f is $p(x) = kx$ for $x = 1, 2, 3, 4$. Since $\sum_x p(x) = 1$:

$$k(1 + 2 + 3 + 4) = 10k = 1 \quad \Rightarrow \quad k = 0.1.$$

2. Expectation and variance:

$$E(X) = \sum x\,p(x) = 0.1(1^2 + 2^2 + 3^2 + 4^2) = 0.1 \times 30 = 3.$$

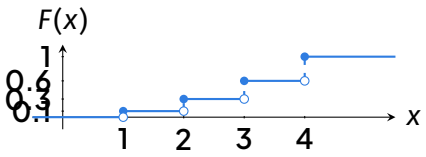
$$E(X^2) = \sum x^2 p(x) = 0.1(1^3 + 2^3 + 3^3 + 4^3) = 0.1 \times 100 = 10.$$

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = 10 - 9 = 1.$$

3. Mode: The probabilities are $p(1) = 0.1$, $p(2) = 0.2$, $p(3) = 0.3$, $p(4) = 0.4$. The maximum is at $x = 4$, so mode = 4.
   Median: Cumulative probabilities: $F(1) = 0.1$, $F(2) = 0.3$, $F(3) = 0.6$, $F(4) = 1$. The smallest $x$ with $F(x) \geq 0.5$ is $x = 3$, so median = 3.

4. Sketch of $F(x)$: Step function with jumps at 1, 2, 3, 4:

$$F(x) = \begin{cases} 0 & x < 1 \\ 0.1 & 1 \leq x < 2 \\ 0.3 & 2 \leq x < 3 \\ 0.6 & 3 \leq x < 4 \\ 1 & x \geq 4 \end{cases}$$




Cumulative distribution function for Exercise 5

**SOLUTION** **Exercise 6 — From the c.d.f to the p.m.f**

1. The p.m.f is obtained from the jumps of $F$:

$$P(X = 1) = F(1) - F(1^-) = 0.2 - 0 = 0.2,$$

$$P(X = 2) = F(2) - F(2^-) = 0.5 - 0.2 = 0.3,$$

$$P(X = 3) = F(3) - F(3^-) = 0.8 - 0.5 = 0.3,$$

$$P(X = 4) = F(4) - F(4^-) = 1 - 0.8 = 0.2.$$

Table:

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $P(X = x)$ | 0.2 | 0.3 | 0.3 | 0.2 |

2. Expectation and variance:

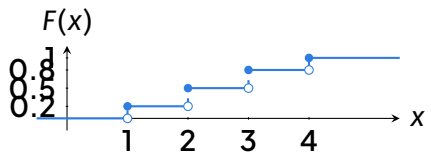
$$E(X) = 1(0.2) + 2(0.3) + 3(0.3) + 4(0.2) = 0.2 + 0.6 + 0.9 + 0.8 = 2.5.$$

$$E(X^2) = 1^2(0.2) + 2^2(0.3) + 3^2(0.3) + 4^2(0.2) = 0.2 + 1.2 + 2.7 + 3.2 = 7.3.$$

$$\mathrm{Var}(X) = 7.3 - (2.5)^2 = 7.3 - 6.25 = 1.05.$$

3. Median: $F(1) = 0.2 < 0.5$, $F(2) = 0.5 \geq 0.5$. The smallest $x$ with $F(x) \geq 0.5$ is $x = 2$, so median = 2.

4. Graph of $F(x)$:




Cumulative distribution function for Exercise 6

**SOLUTION** **Exercise 7 — A fair game?**

1. Net gain $G$ = winnings − 100.

| $g$ | 400 | 100 | −100 |
|---|---|---|---|
| $P(G = g)$ | $\frac{1}{10} = 0.1$ | $\frac{1}{5} = 0.2$ | $1 - 0.1 - 0.2 = 0.7$ |

2. $E(G) = 400(0.1) + 100(0.2) + (-100)(0.7) = 40 + 20 - 70 = -10.$

$E(G^2) = 400^2(0.1) + 100^2(0.2) + (-100)^2(0.7) = 16000 + 2000 + 7000 = 25000.$

$$\mathrm{Var}(G) = 25000 - (-10)^2 = 24900, \quad \sigma_G = \sqrt{24900} \approx 157.8.$$

3. The game is not fair because $E(G) = -10 \neq 0$ (the player loses on average 10 FCFA). For a fair game, the stake $s$ must satisfy $E(\text{winnings}) - s = 0$. $E(\text{winnings}) = 500(0.1) + 200(0.2) = 50 + 40 = 90$. Hence a fair stake is 90 FCFA.

**SOLUTION** **Exercise 8 — Linear combinations of independent variables**

$X$ and $Y$ independent, $E(X) = 2$, $\mathrm{Var}(X) = 1$, $E(Y) = 5$, $\mathrm{Var}(Y) = 4$.

1. $E(2X + 3Y) = 2E(X) + 3E(Y) = 2(2) + 3(5) = 4 + 15 = 19$.
2. $\mathrm{Var}(2X + 3Y) = 2^2\,\mathrm{Var}(X) + 3^2\,\mathrm{Var}(Y) = 4(1) + 9(4) = 4 + 36 = 40$. Independence is used to assert $\mathrm{Cov}(X, Y) = 0$, so the variance of the sum is the sum of the variances (no covariance term).
3. $E(X - Y) = E(X) - E(Y) = 2 - 5 = -3$. $\mathrm{Var}(X - Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) = 1 + 4 = 5$ (again using independence to drop the covariance term).

**SOLUTION** **Exercise 9 — Moment generating function of a Bernoulli variable (10 marks)**

1. (3 marks) $X \sim \text{Bernoulli}(p)$: $P(X = 1) = p$, $P(X = 0) = 1 - p$.

$$M_X(t) = E(e^{tX}) = (1 - p)e^{t\cdot 0} + p\,e^{t\cdot 1} = 1 - p + pe^t.$$

2. (4 marks) Differentiate:

$$M_X'(t) = pe^t, \quad M_X''(t) = pe^t.$$

Then $E(X) = M_X'(0) = p$. $E(X^2) = M_X''(0) = p$ (since $X^2 = X$ for Bernoulli).

3. (3 marks) $\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = p - p^2 = p(1 - p)$.

Mark scheme: 3 marks for correct MGF derivation; 4 marks for correct differentiation and evaluation at 0; 3 marks for correct variance derivation. Examiner expects clear statement of $M_X(t)$, correct differentiation, and explicit use of $M_X'(0) = E(X)$, $M_X''(0) = E(X^2)$.

**SOLUTION** **Exercise 10 — Recovering a distribution from its moments (12 marks)**

1. (3 marks) Probabilities $a, b, c$ for $x = 1, 2, 3$:

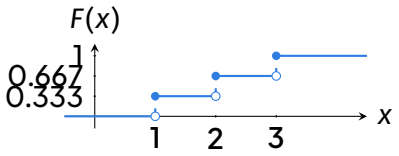
$$\begin{cases} a + b + c = 1 & \text{(sum of probabilities)} \\ a + 2b + 3c = 2 & (E(X) = 2) \\ a + 4b + 9c = \frac{14}{3} & (E(X^2) = \frac{14}{3}) \end{cases}$$

2. (4 marks) Subtract first from second: $b + 2c = 1$. Subtract second from third: $2b + 6c = \frac{8}{3} \Rightarrow b + 3c = \frac{4}{3}$. Subtract: $c = \frac{1}{3}$. Then $b = 1 - 2c = \frac{1}{3}$, $a = 1 - b - c = \frac{1}{3}$. So $a = b = c = \frac{1}{3}$.

3. (2 marks) All probabilities equal $\frac{1}{3}$. Mode: every value 1, 2, 3 is a mode (or the distribution has no unique mode). Median: $F(1) = \frac{1}{3} < 0.5$, $F(2) = \frac{2}{3} \geq 0.5 \Rightarrow$ median = 2.

4. (3 marks) CDF:

$$F(x) = \begin{cases} 0 & x < 1 \\ \frac{1}{3} & 1 \leq x < 2 \\ \frac{2}{3} & 2 \leq x < 3 \\ 1 & x \geq 3 \end{cases}$$




Cumulative distribution function for Exercise 10

Mark scheme: 3 marks for setting up three correct equations; 4 marks for solving correctly; 2 marks for correct mode and median; 3 marks for correct CDF and sketch. Examiner expects clear algebraic steps and recognition of uniform distribution.

**SOLUTION** **Exercise 11 — Defective items in a sample — GCE style** **(16 marks)**

1. (2 marks) Sum of probabilities = 1:

$$0.512 + k + 0.096 + 0.008 = 1 \;\Rightarrow\; 0.616 + k = 1 \;\Rightarrow\; k = 0.384.$$

2. (5 marks)

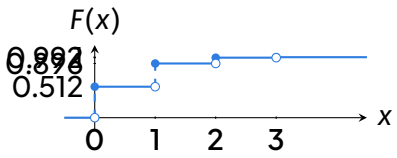
$$E(X) = 0(0.512) + 1(0.384) + 2(0.096) + 3(0.008) = 0.384 + 0.192 + 0.024 = 0.6.$$

$$E(X^2) = 0^2(0.512) + 1^2(0.384) + 4(0.096) + 9(0.008) = 0.384 + 0.384 + 0.072 = 0.84.$$

$$\text{Var}(X) = 0.84 - 0.6^2 = 0.84 - 0.36 = 0.48.$$

3. (4 marks) CDF:

$$F(x) = \begin{cases} 0 & x < 0 \\ 0.512 & 0 \le x < 1 \\ 0.896 & 1 \le x < 2 \\ 0.992 & 2 \le x < 3 \\ 1 & x \ge 3 \end{cases}$$




Cumulative distribution function for Exercise 11

4. (2 marks) Mode: $x = 0$ (probability 0.512 is largest). Median: $F(0) = 0.512 \ge 0.5 \Rightarrow$ median = 0.

5. (3 marks) MGF:

$$M_X(t) = 0.512e^{0} + 0.384e^{t} + 0.096e^{2t} + 0.008e^{3t}.$$

$$M_X'(t) = 0.384e^{t} + 0.192e^{2t} + 0.024e^{3t}.$$

$$M_X'(0) = 0.384 + 0.192 + 0.024 = 0.6 = E(X). \quad \textbf{(Verified)}$$

Mark scheme: 2 marks for $k$; 5 marks for $E(X)$ and $\mathrm{Var}(X)$ (2 for $E(X)$, 2 for $E(X^2)$, 1 for Var); 4 marks for CDF (2 for correct values, 2 for sketch); 2 marks for mode and median; 3 marks for MGF and verification. Examiner expects clear arithmetic, correct CDF definition, and proper use of MGF to find $E(X)$.

# 12 SUMMARY SHEET

# 13 DISCRETE RANDOM VARIABLES

## 13.1 Discrete random variables and probability distributions

**DEFINITION** **Discrete random variable**

A **discrete random variable** (DRV) $X$ is a function that assigns a real number to each outcome in a sample space $\Omega$, such that the set of possible values $\{x_1, x_2, \dots\}$ is either finite or countably infinite. We write $X : \Omega \to \mathbb{R}$.

**EXEMPLE** **Examples of discrete random variables**

- Number of heads when tossing a coin three times: $X \in \{0, 1, 2, 3\}$.
- Score on a fair six-sided die: $X \in \{1, 2, 3, 4, 5, 6\}$.
- Number of customers arriving at a shop in one hour: $X \in \{0, 1, 2, \dots\}$ (countably infinite).
- Number of defective items in a batch of 20: $X \in \{0, 1, \dots, 20\}$.

**DEFINITION** **Probability mass function (p.m.f.)**

The **probability mass function** (p.m.f.) of a discrete random variable $X$ is the function $p(x) = P(X = x)$ defined for all $x \in \mathbb{R}$. It satisfies:

1. $p(x_i) \geq 0$ for every possible value $x_i$,
2. $\sum_{\text{all } x_i} p(x_i) = 1$.

The collection of pairs $(x_i, p(x_i))$ is called the **probability distribution** of $X$.

**PROPERTY** **Probability distribution table**

The probability distribution of a DRV is conventionally presented as a table:

| $x_i$ | $x_1$ | $x_2$ | $\cdots$ | $x_n$ |
|---|---|---|---|---|
| $P(X = x_i)$ | $p_1$ | $p_2$ | $\cdots$ | $p_n$ |

with $\sum_{i=1}^{n} p_i = 1$. Always verify this sum as a validity check.

**METHOD** **Constructing a probability distribution**

1. Identify the sample space $\Omega$ of the random experiment.
2. Define the random variable $X$ and list all its possible values $x_1, x_2, \ldots$.
3. Compute $p(x_i) = P(X = x_i)$ for each value using the rules of probability.
4. Present the distribution in a table and verify $\sum p(x_i) = 1$.

**WORKED EXAMPLE 16** **Constructing a distribution**

A fair die is rolled twice. Let $X$ be the maximum of the two scores. Find the probability distribution of $X$.

**SOLUTION**

Solution. The sample space has $6 \times 6 = 36$ equally likely outcomes. $X$ takes values $1, 2, 3, 4, 5, 6$.

- $P(X = 1) = P(\text{both scores are } 1) = \frac{1}{36}$.
- $P(X = 2) = P(\max = 2) = P(\text{both} \le 2) - P(\text{both} \le 1) = \frac{4}{36} - \frac{1}{36} = \frac{3}{36}$.
- Similarly: $P(X = 3) = \frac{5}{36}$, $P(X = 4) = \frac{7}{36}$, $P(X = 5) = \frac{9}{36}$, $P(X = 6) = \frac{11}{36}$.

Distribution table:

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $P(X = x)$ | $\frac{1}{36}$ | $\frac{3}{36}$ | $\frac{5}{36}$ | $\frac{7}{36}$ | $\frac{9}{36}$ | $\frac{11}{36}$ |

Check: $\frac{1+3+5+7+9+11}{36} = \frac{36}{36} = 1$. ✓

**EXERCISE 16** **Finding an unknown constant**

A discrete random variable $X$ has probability distribution given by

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $P(X = x)$ | $k$ | $2k$ | $3k$ | $2k$ | $k$ |

Find the value of $k$ and hence write down the full distribution.

**SOLUTION** **Exercise 1**

$\sum P(X = x) = k + 2k + 3k + 2k + k = 9k = 1 \implies k = \frac{1}{9}$. Distribution:

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $P(X = x)$ | $\frac{1}{9}$ | $\frac{2}{9}$ | $\frac{3}{9}$ | $\frac{2}{9}$ | $\frac{1}{9}$ |

## 13.2 The probability mass function and expectation

**DEFINITION** **Expectation (mean) of a discrete random variable**

The **expectation** or **mean** of a discrete random variable $X$, denoted $E(X)$ or $\mu$, is

$$E(X) = \sum_{\text{all } x_i} x_i \, P(X = x_i) = \sum_i x_i p_i,$$

provided the series converges absolutely.

**PROPERTY** **Expectation of a function of $X$**

For any real-valued function $g$,

$$E[g(X)] = \sum_i g(x_i)\, p(x_i).$$

In particular, $E(X^2) = \sum_i x_i^2 p_i$.

**PROPERTY** **Linearity of expectation**

For any constants $a, b \in \mathbb{R}$ and any discrete random variables $X, Y$ (independent or not):

$$E(aX + b) = aE(X) + b,$$
$$E(X + Y) = E(X) + E(Y).$$

EXEMPLE **Computing $E(X)$ and $E(X^2)$**

Using the distribution from the previous worked example (maximum of two dice):

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 | $xp(x)$ | $x^2p(x)$ |
|---|---|---|---|---|---|---|---|---|
| $p(x)$ | $\frac{1}{36}$ | $\frac{3}{36}$ | $\frac{5}{36}$ | $\frac{7}{36}$ | $\frac{9}{36}$ | $\frac{11}{36}$ | $\frac{161}{36}$ | $\frac{791}{36}$ |

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